

11.7.4.38 (ID 1618) Fire Mode Reverse

Location in the menu: **P3.5.1.40**

Use this parameter to select the digital input signal that gives a command for reverse rotation direction during the Fire Mode. This function has no effect in normal operation.

11.7.4.39 (ID 1715) Auto-Cleaning Activation

Location in the menu: **P3.5.1.41**

Use this parameter to select the digital input signal that starts the Auto-cleaning. The Auto-cleaning stops if the activation signal is removed before the process is complete.

NOTICE

If the input is activated, the drive starts.

11.7.4.40 (ID 426) Pump 1 Interlock

Location in the menu: **P3.5.1.42**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.41 (ID 427) Pump 2 Interlock

Location in the menu: **P3.5.1.43**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.42 (ID 428) Pump 3 Interlock

Location in the menu: **P3.5.1.44**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.43 (ID 429) Pump 4 Interlock

Location in the menu: **P3.5.1.45**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.44 (ID 430) Pump 5 Interlock

Location in the menu: **P3.5.1.46**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.45 (ID 486) Pump 6 Interlock

Location in the menu: **P3.5.1.47**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.46 (ID 487) Pump 7 Interlock

Location in the menu: **P3.5.1.48**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.47 (ID 488) Pump 8 Interlock

Location in the menu: **P3.5.1.49**

Use this parameter to select the digital input signal that is used as interlock signal for the Multi-pump system. The selection of the application with parameter P1.2 Application gives the default value.

11.7.4.48 (ID 1053) Reset kWh Trip Counter

Location in the menu: **P3.5.1.52**

Use this parameter to select the digital input signal that resets the kWh Trip Counter.

11.7.4.49 (ID 496) Parameter Set 1/2 Selection

Location in the menu: **P3.5.1.53**

Use this parameter to set the digital input that selects the parameter set to be used. This function is enabled if any other slot than 'DigIN Slot0' is selected to this parameter. The parameter set selection is allowed only when the drive is stopped.

- Contact Open = Parameter Set 1 is loaded as the active set
- Contact Closed = Parameter Set 2 is loaded as the active set

Parameter values are stored to Set 1 and Set 2 by parameters B6.5.4 Save to Set 1 and B6.5.4 Save to Set 2. These parameters can be used either from keypad or VACON® Live PC tool.

11.7.4.50 (ID 15576) Multi-setpoint Selection 0

Location in the menu: **P3.5.1.54**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.7.4.51 (ID 15577) Multi-setpoint Selection 1

Location in the menu: **P3.5.1.55**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.7.4.52 (ID 15578) Multi-setpoint Selection 2

Location in the menu: **P3.5.1.56**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.7.4.53 (ID 15579) Multi-setpoint Selection 3

Location in the menu: **P3.5.1.57**

Use this parameter to select the digital input signal that is used as a selector for the PID Multi-setpoint function.

11.7.4.54 (ID 15590) Flow Switch

Location in the menu: **P3.5.1.58**

Use this parameter to select the digital input signal that is the feedback signal from the Flow Switch.

11.7.4.55 (ID 15513) AHF Over Temperature

- Location in the menu: **P3.5.1.59** (In I/O Configuration parameter menu)
- Location in the menu: **P3.23.3** (In Advanced Harmonic Filter parameter menu)

Use this parameter to set the digital input signal that activates AHF Over Temp (fault ID 1118).

11.7.5 Analog Inputs

11.7.5.1 (ID 377) AI1 Signal Selection

Location in the menu: **P3.5.2.1.1**

Use this parameter to connect the AI signal to the analog input of your selection.

This parameter is programmable. See [11.7.3 Default Functions of Programmable Inputs](#).

11.7.5.2 (ID 378) AI1 Signal Filter Time

Location in the menu: **P3.5.2.1.2**

Use this parameter to filter out disturbances in the analog input signal. To activate this parameter, enter a value greater than 0.

N O T I C E

A long filter time makes the regulation response slow.

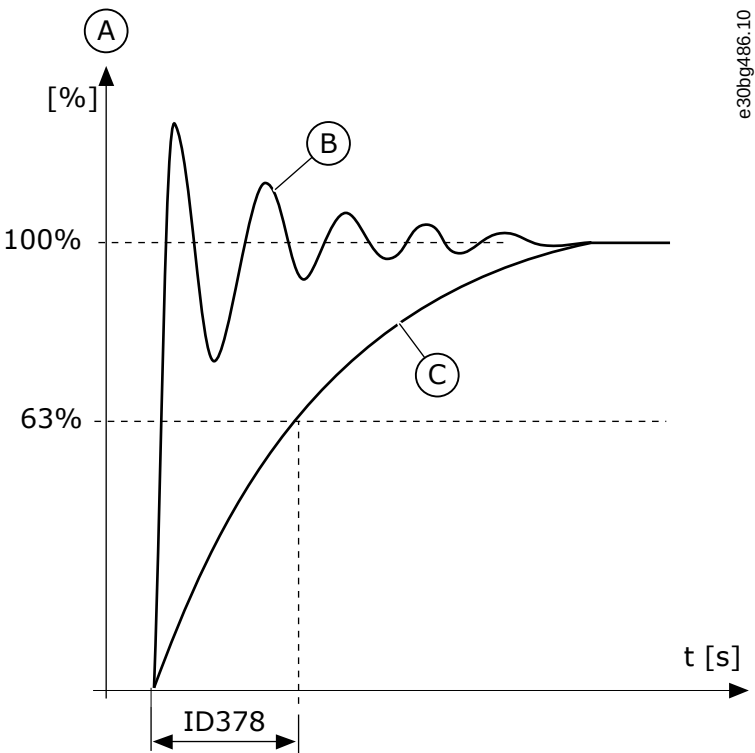


Illustration 56: The AI1 Signal Filtering

A	Analog input signal	C	Filtered signal
B	Unfiltered signal		

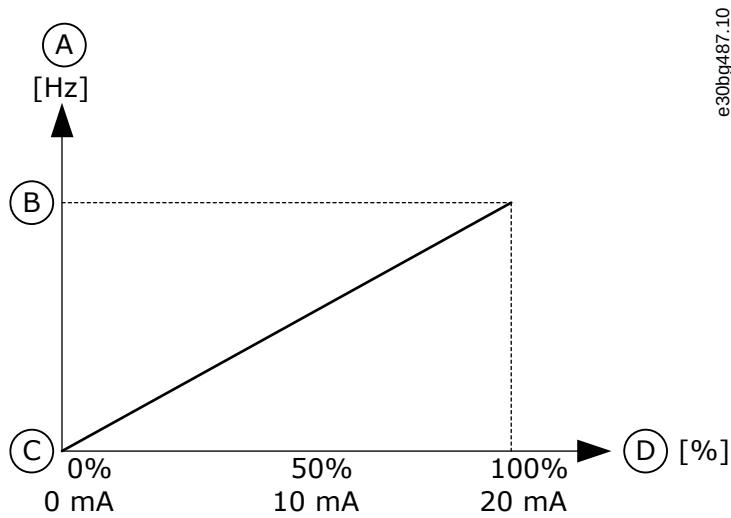
11.7.5.3 (ID 379) AI1 Signal Range

Location in the menu: **P3.5.2.1.3**

Use this parameter to change the range of the analog signal. The value of this parameter is bypassed if the custom scaling parameters are used.

Use the DIP switches on the control board to set the type of the analog input signal (current or voltage). For more information, see the Installation manual. It is also possible to use the analog input signal as frequency reference. The selection of the value 0 or 1 change the scaling of the analog input signal.

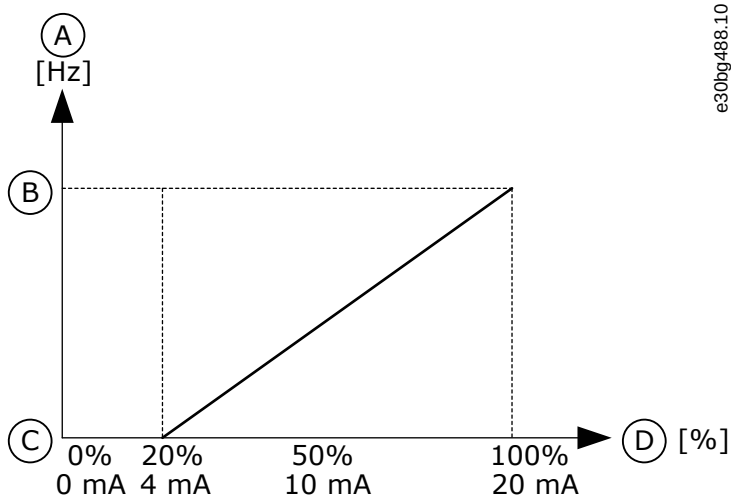
Selection number	Selection name	Description
0	0...10 V/0...20 mA	The range of the analog input signal is 0...10 V or 0...20 mA (the DIP switch settings on the control board tell which one). The input signal is 0...100%.
1	2...10 V/4...20 mA	The range of the analog input signal is 2...10 V or 4...20 mA (the DIP switch settings on the control board tell which one). The input signal is 20...100%.



e30bg487.10

Illustration 57: The Analog Input Signal Range, Selection 0

A	Frequency reference	C	Min freq reference
B	Max freq reference	D	Analog input signal



e30bg488.10

Illustration 58: The Analog Input Signal Range, Selection 1

A	Frequency reference	C	Min freq reference
B	Max freq reference	D	Analog input signal

11.7.5.4 (ID 380) AI1 Custom. MIN

Location in the menu: **P3.5.2.1.4**

Use this parameter to adjust the range of the analog input signal between -160% and 160%.

11.7.5.5 (ID 381) AI1 Custom. MAX

Location in the menu: **P3.5.2.1.5**

Use this parameter to adjust the range of the analog input signal between -160% and 160%.

Use the analog input signal, for example, as frequency reference, and set the parameters P3.5.2.1.4 and P3.5.2.1.5 between 40 and 80%. In these conditions, the frequency reference changes between the Minimum frequency reference and the Maximum frequency reference, and the analog input signal changes between 8 and 16 mA.

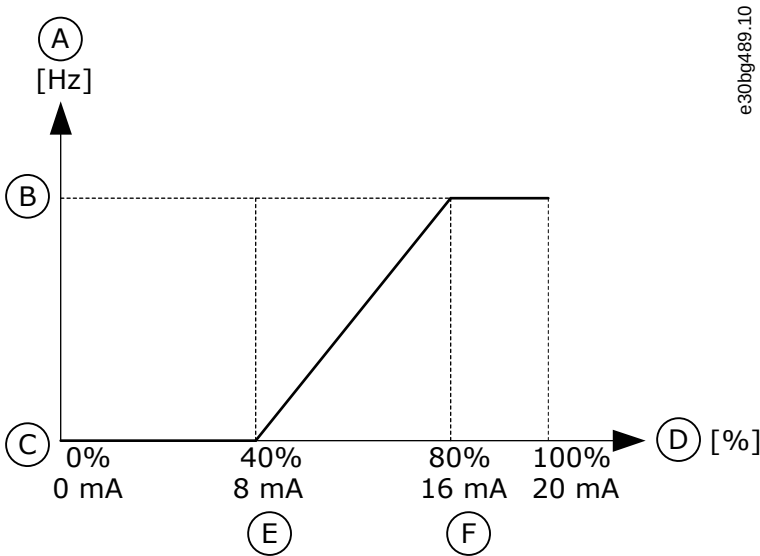


Illustration 59: AI1 Signal Custom. Min/Max

A	Frequency reference	D	Analog input signal
B	Max freq reference	E	AI custom min
C	Min freq reference	F	AI custom max

11.7.5.6 (ID 387) AI1 Signal Inversion

Location in the menu: **P3.5.2.1.6**

Use this parameter to invert the analog input signal. When the analog input signal is inverted, the curve of the signal becomes the opposite.

It is possible to use the analog input signal as frequency reference. The selection of the value 0 or 1 change scaling of the analog input signal.

Selection number	Selection name	Description
0	Normal	No inversion. The value 0% of the analog input signal agrees to the Minimum Frequency Reference. The value 100% of the analog input signal agrees to the Maximum Frequency Reference.
1	Inverted	Signal inversion. The value 0% of the analog input signal agrees to the Maximum Frequency Reference. The value 100% of the analog input signal agrees to the Minimum Frequency Reference.

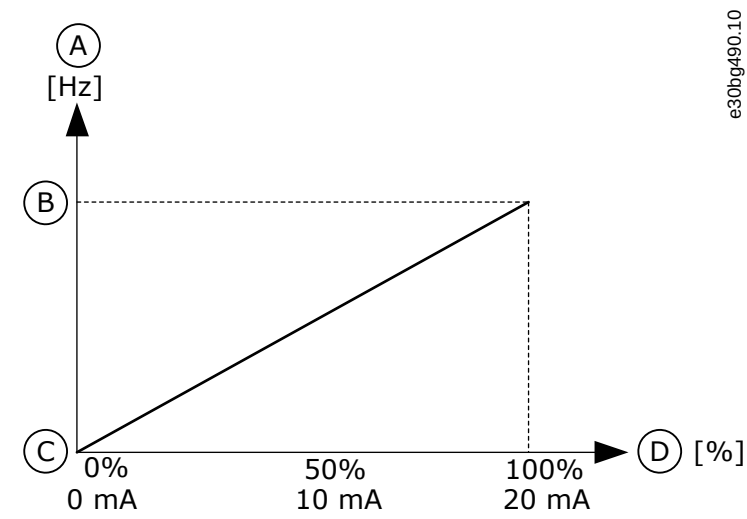


Illustration 60: AI1 Signal Inversion, Selection 0

A	Frequency reference	C	Minimum frequency reference
B	Maximum frequency reference	D	Analog input signal

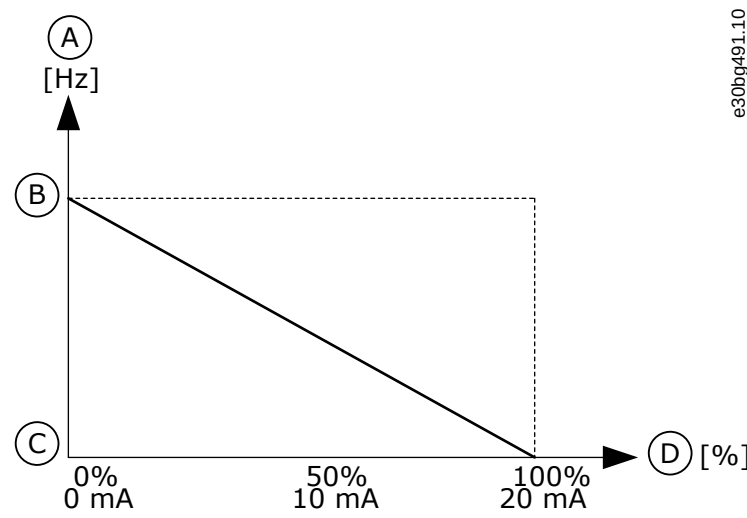


Illustration 61: AI1 Signal Inversion, Selection 1

A	Frequency reference	C	Minimum frequency reference
B	Maximum frequency reference	D	Analog input signal

11.7.6 Digital Outputs

11.7.6.1 (ID 11001) RO1 Function

Location in the menu: **P3.5.3.2.1**

Use this parameter to select a function or a signal that is connected to the relay output.

Selection number	Selection name	Description
0	Not used	The output is not used.
1	Ready	The AC drive is ready to operate.
2	Run	The AC drive operates (the motor runs).
3	General fault	A fault trip occurred.
4	General fault inverted	A fault trip did not occur.
5	General alarm	An alarm occurred.
6	Reversed	The reverse command is given.
7	At speed	The output frequency has become the same as the set frequency reference.
8	Thermistor fault	A thermistor fault occurred.
9	Motor regulator activated	One of the limit regulators (for example current limit or torque limit) is activated.
10	Start signal active	The start command of the drive is active.
11	Keypad control active	The selection is keypad control (the active control place is keypad).
12	I/O control B active	The selection is I/O control place B (the active control place is I/O B).
13	Limit supervision 1	The limit supervision activates if the signal value goes below or above the set supervision limit (P3.8.3 or P3.8.7).
14	Limit supervision 2	The limit supervision activates if the signal value goes below or above the set supervision limit (P3.8.3 or P3.8.7).
15	Fire mode active	The Fire mode function is active.
16	Flushing active	The Flushing function is active.
17	Preset Frequency active	The selection of preset frequency was made with digital input signals.
18	Quick Stop active	The Quick stop function is activated.
19	PID in Sleep mode	The PID controller is in the sleep mode.
20	PID Soft Fill activated	The Soft fill function of the PID controller is activated.
21	PID feedback supervision	The feedback value of the PID controller is not in the supervision limits.
22	ExtPID feedback supervision	The External PID controller feedback value is not in the supervision limits.
23	Input pressure alarm	The input pressure of the pump is below the value that was set with parameter P3.13.9.7.
24	Frost protection alarm	The measured temperature of the pump is below the level that was set with parameter P3.13.10.5.
25	Time channel 1	The status of Time channel 1.
26	Time channel 2	The status of Time channel 2.
27	Time channel 3	The status of Time channel 3.
28	Fieldbus Control Word bit 13	The digital (relay) output control from the Fieldbus control word bit 13.
29	Fieldbus Control Word bit 14	The digital (relay) output control from the Fieldbus control word bit 14.

Selection number	Selection name	Description
30	Fieldbus Control Word bit 15	The digital (relay) output control from the Fieldbus control word bit 15.
31	Fieldbus Process Data In1 bit 0	The digital (relay) output control from the Fieldbus Process Data In1, bit 0.
32	Fieldbus Process Data In1 bit 1	The digital (relay) output control from the Fieldbus Process Data In1, bit 1.
33	Fieldbus Process Data In1 bit 2	The digital (relay) output control from the Fieldbus Process Data In1, bit 2.
34	Maintenance counter 1 alarm	The maintenance counter reached the alarm limit that is set with parameter P3.16.2.
35	Maintenance counter 1 fault	The maintenance counter reached the alarm limit that is set with parameter P3.16.3.
36	Block Out.1	The output of the programmable Block 1. See parameter menu M3.19 Drive customizer.
37	Block Out.2	The output of the programmable Block 2. See parameter menu M3.19 Drive customizer.
38	Block Out.3	The output of the programmable Block 3. See parameter menu M3.19 Drive customizer.
39	Block Out.4	The output of the programmable Block 4. See parameter menu M3.19 Drive customizer.
40	Block Out.5	The output of the programmable Block 5. See parameter menu M3.19 Drive customizer.
41	Block Out.6	The output of the programmable Block 6. See parameter menu M3.19 Drive customizer.
42	Block Out.7	The output of the programmable Block 7. See parameter menu M3.19 Drive customizer.
43	Block Out.8	The output of the programmable Block 8. See parameter menu M3.19 Drive customizer.
44	Block Out.9	The output of the programmable Block 9. See parameter menu M3.19 Drive customizer.
45	Block Out.10	The output of the programmable Block 10. See parameter menu M3.19 Drive customizer.
46	Jockey pump control	The control signal for the external jockey pump.
47	Priming pump control	The control signal for the external priming pump.
48	Auto-cleaning active	The Pump auto-cleaning function is activated.
49	Multi-pump K1 control	The contactor control for the Multi-pump function.
50	Multi-pump K2 control	The contactor control for the Multi-pump function.
51	Multi-pump K3 control	The contactor control for the Multi-pump function.
52	Multi-pump K4 control	The contactor control for the Multi-pump function.
53	Multi-pump K5 control	The contactor control for the Multi-pump function.
54	Multi-pump K6 control	The contactor control for the Multi-pump function.

Selection number	Selection name	Description
55	Multi-pump K7 control	The contactor control for the Multi-pump function.
56	Multi-pump K8 control	The contactor control for the Multi-pump function.
69	Selected parameter set	Shows the active parameter set: OPEN = Parameter set 1 active CLOSED = Parameter set 2 active
72	AHF Capacitor Disconnection	
73	AHF Capacitor Disconnection Inverted	
74	Run Indication	
75	Motor Switch Open	The Motor Switch function has detected that the switch between the drive and the motor is open.

11.7.6.2 (ID 11002) RO1 ON Delay

Location in the menu: **P3.5.3.2.2**

Use this parameter to set the ON delay for the relay output.

11.7.6.3 (ID 11003) RO1 OFF Delay

Location in the menu: **P3.5.3.2.3**

Use this parameter to set the OFF delay for the relay output.

11.7.7 Analog Outputs

11.7.7.1 (ID 10050) AO1 Function

Location in the menu: **P3.5.4.1.1**

Use this parameter to select a function or a signal that is connected to the analog output. The contents of the analog output signal 1 are specified in this parameter. The scaling of the analog output signal depends on the signal.

Selection number	Selection name	Description
0	Test 0% (Not used)	The analog output is set to 0% or 20% so that it agrees with parameter P3.5.4.1.3.
1	TEST 100%	The analog output is set to 100% of the signal (10 V/20 mA).
2	Output frequency	The actual output frequency from 0 to Maximum frequency reference.
3	Frequency reference	The actual frequency reference from 0 to Maximum frequency reference.
4	Motor speed	The actual motor speed from 0 to Motor nominal speed.
5	Output current	The output current of the drive from 0 to Motor nominal current.
6	Motor torque	The actual motor torque from 0 to motor nominal torque (100%).
7	Motor power	The actual motor power from 0 to Motor nominal power (100%).
8	Motor voltage	The actual motor voltage from 0 to Motor nominal voltage.
9	DC link voltage	The actual DC link voltage 0...1000 V.
10	PID Setpoint	The actual setpoint value of the PID Controller (0...100%).

Selection number	Selection name	Description
11	PID Feedback	The actual feedback value of the PID Controller (0...100%).
12	PID output	The output of the PID controller (0...100%).
13	ExtPID output	The External PID controller output (0...100%).
14	Fieldbus Process Data In 1	Fieldbus Process Data In 1: 0...10000 (this agrees with 0... 100.00%).
15	Fieldbus Process Data In 2	Fieldbus Process Data In 2: 0...10000 (this agrees with 0... 100.00%).
16	Fieldbus Process Data In 3	Fieldbus Process Data In 3: 0...10000 (this agrees with 0... 100.00%).
17	Fieldbus Process Data In 4	Fieldbus Process Data In 4: 0...10000 (this agrees with 0... 100.00%).
18	Fieldbus Process Data In 5	Fieldbus Process Data In 5: 0...10000 (this agrees with 0... 100.00%).
19	Fieldbus Process Data In 6	Fieldbus Process Data In 6: 0...10000 (this agrees with 0... 100.00%).
20	Fieldbus Process Data In 7	Fieldbus Process Data In 7: 0...10000 (this agrees with 0... 100.00%).
21	Fieldbus Process Data In 8	Fieldbus Process Data In 8: 0...10000 (this agrees with 0... 100.00%).
22	Block Out.1	The output of the programmable Block 1: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
23	Block Out.2	The output of the programmable Block 2: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
24	Block Out.3	The output of the programmable Block 3: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
25	Block Out.4	The output of the programmable Block 4: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
26	Block Out.5	The output of the programmable Block 5: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
27	Block Out.6	The output of the programmable Block 6: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
28	Block Out.7	The output of the programmable Block 7: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
29	Block Out.8	The output of the programmable Block 8: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
30	Block Out.9	The output of the programmable Block 9: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.
31	Block Out.10	The output of the programmable Block 10: 0...10000 (this agrees with 0...100.00%). See parameter menu M3.19 Drive customizer.

11.7.7.2 (ID 10051) AO1 Filter Time

Location in the menu: **P3.5.4.1.2**

Use this parameter to set the filtering time for the analog signal. The filtering function is disabled when the filtering time is 0. See P3.5.2.1.2.

11.7.7.3 (ID 10052) AO1 Minimum

Location in the menu: **P3.5.4.1.3**

Use this parameter to change the range of the analog output signal. For example, if '4 mA' is selected, the range of analog output signal is 4...20 mA. Select the signal type (current/voltage) with the DIP switches. The analog output scaling is different in P3.5.4.1.4. See also P3.5.2.1.3.

11.7.7.4 (ID 10053) AO1 Minimum Scale

Location in the menu: **P3.5.4.1.4**

Use this parameter to scale the analog output signal. The scaling values (min and max) are given in the process unit specified by the selection of the AO function.

11.7.7.5 (ID 10054) AO1 Maximum Scale

Location in the menu: **P3.5.4.1.5**

Use this parameter to scale the analog output signal. The scaling values (min and max) are given in the process unit specified by the selection of the AO function.

Select, for example, the output frequency of the drive for the contents of the analog output signal, and set parameters P3.5.4.1.4 and P3.5.4.1.5 between 10 and 40 Hz. Then the output frequency of the drive changes between 10 and 40 Hz, and the analog output signal changes between 0 and 20 mA.

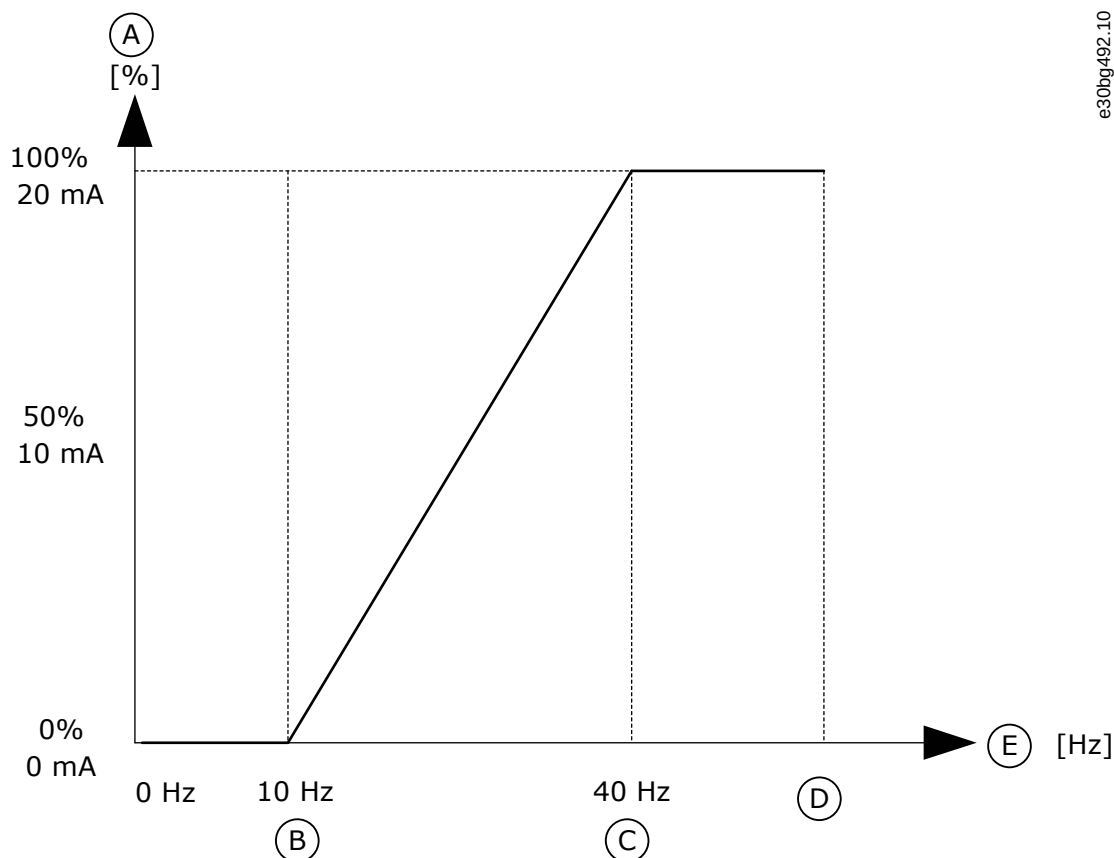


Illustration 62: Scaling of the AO1 Signal

A	Analog output signal	D	Maximum frequency reference
B	Analog output minimum scale	E	Output frequency
C	Analog output maximum scale		

11.8 Fieldbus Data

11.8.1 (ID 852) FB DataOut 1 Selection

Location in the menu: **P3.6.1**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.2 (ID 853) FB DataOut 2 Selection

Location in the menu: **P3.6.2**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.3 (ID 854) FB DataOut 3 Selection

Location in the menu: **P3.6.3**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.4 (ID 855) FB DataOut 4 Selection

Location in the menu: **P3.6.4**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.5 (ID 856) FB DataOut 5 Selection

Location in the menu: **P3.6.5**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.6 (ID 857) FB DataOut 6 Selection

Location in the menu: **P3.6.6**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.7 (ID 858) FB DataOut 7 Selection

Location in the menu: **P3.6.7**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.8 (ID 859) FB DataOut 8 Selection

Location in the menu: **P3.6.8**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.9 (ID 890) FB DataOut 9 Selection

Location in the menu: **P3.6.9**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.10 (ID 891) FB DataOut 10 Selection

Location in the menu: **P3.6.10**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.11 (ID 892) FB DataOut 11 Selection

Location in the menu: **P3.6.11**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.12 (ID 893) FB DataOut 12 Selection

Location in the menu: **P3.6.12**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.13 (ID 894) FB DataOut 13 Selection

Location in the menu: **P3.6.13**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.14 (ID 895) FB DataOut 14 Selection

Location in the menu: **P3.6.14**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.15 (ID 896) FB DataOut 15 Selection

Location in the menu: **P3.6.15**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.16 (ID 897) FB DataOut 16 Selection

Location in the menu: **P3.6.16**

Use this parameter to select the data that is sent to the fieldbus with the ID number of the parameter or monitor value.

The data is scaled to an unsigned 16-bit format according to the format on the control panel. For example, value 25.5 on the display equals 255.

11.8.17 (ID 888) Fieldbus Sign of Actual

Location in the menu: **P3.6.17**

Use this parameter to select the behavior for the sign of the fieldbus Speed Actual.

Selection number	Selection name	Description
0	Positive	The sign of the actual is always positive. The running direction is visible from the status word bit 2. A TRUE value corresponds to reversing actual speed.
1	Actual direction	The sign of the actual follows the running direction. Forward is positive, reverse is negative. A TRUE value for status word bit 2 indicates if the direction of the actual differs from the direction requested from FB control commands.
2	Follow reference	A negative sign of the actual indicates if the direction of the actual differs from the direction requested from FB control commands. The status word bit 2 corresponds to the direction requested from FB control commands. A TRUE value corresponds to requested reversing.

11.9 Prohibit Frequencies

11.9.1 The Prohibit Frequencies Function

In some processes, it can be necessary to avoid some frequencies because they make problems of mechanical resonance. With the Prohibit frequencies function, it is possible to prevent the usage of these frequencies. When the input frequency reference increases, the internal frequency reference stays at the low limit, until the input frequency reference is above the high limit.

11.9.2 (ID 509) Prohibit Frequency Range 1 Low Limit

Location in the menu: **P3.7.1**

Use this parameter to prevent the drive operating on the prohibited frequencies.

In some processes, it can be necessary to avoid some frequencies because they cause mechanical resonance.

11.9.3 (ID 510) Prohibit Frequency Range 1 High Limit

Location in the menu: **P3.7.2**

Use this parameter to prevent the drive operating on the prohibited frequencies. In some processes, it can be necessary to avoid some frequencies because they cause mechanical resonance.

11.9.4 (ID 511) Prohibit Frequency Range 2 Low Limit

Location in the menu: **P3.7.3**

Use this parameter to prevent the drive operating on the prohibited frequencies. In some processes, it can be necessary to avoid some frequencies because they cause mechanical resonance.

11.9.5 (ID 512) Prohibit Frequency Range 2 High Limit

Location in the menu: **P3.7.4**

Use this parameter to prevent the drive operating on the prohibited frequencies. In some processes, it can be necessary to avoid some frequencies because they cause mechanical resonance.

11.9.6 (ID 513) Prohibit Frequency Range 3 Low Limit

Location in the menu: **P3.7.5**

Use this parameter to prevent the drive operating on the prohibited frequencies. In some processes, it can be necessary to avoid some frequencies because they cause mechanical resonance.

11.9.7 (ID 514) Prohibit Frequency Range 3 High Limit

Location in the menu: **P3.7.6**

Use this parameter to prevent the drive operating on the prohibited frequencies. In some processes, it can be necessary to avoid some frequencies because they cause mechanical resonance.

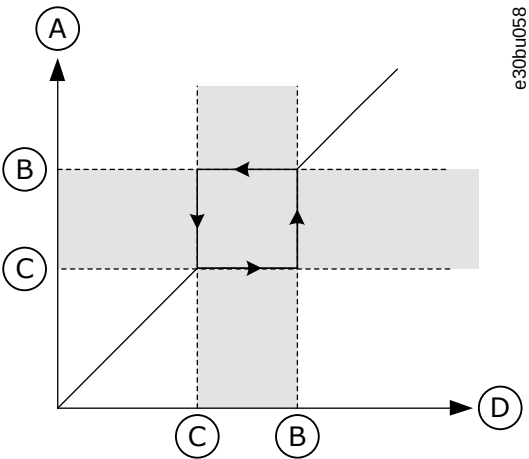


Illustration 63: The Prohibited Frequencies

A	Actual Reference	C	Low Limit
B	High Limit	D	Requested Reference

11.9.8 (ID 518) Ramp Time Factor

Location in the menu: **P3.7.7**

Use this parameter to set the multiplier of the selected ramp times when the output frequency of the drive is between the prohibited frequency limits. The Ramp Time Factor sets the acceleration and the deceleration time when the output frequency is in a prohibited frequency range. The value of the Ramp Time Factor is multiplied by the value of P3.4.1.2 (Acceleration Time 1) or P3.4.1.3 (Deceleration Time 1). For example, the value 0.1 makes the acceleration/deceleration time ten times shorter.

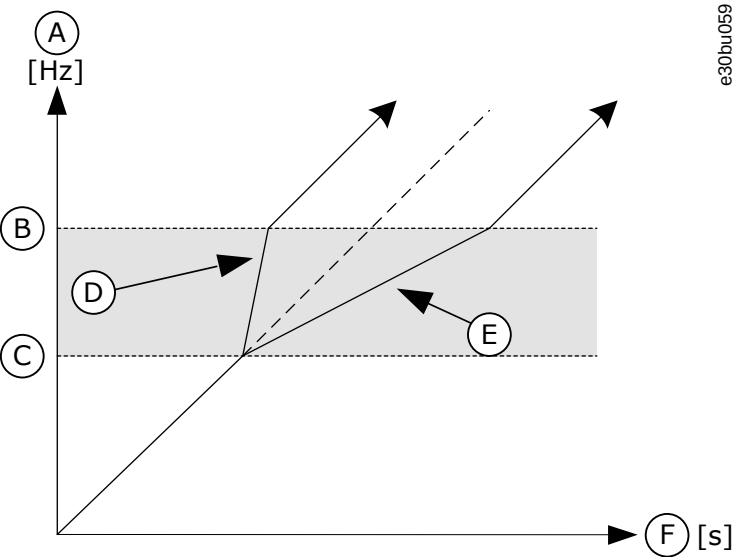


Illustration 64: The Parameter Ramp Time Factor

A	Output Frequency	D	Ramp Time Factor = 0.3
B	High Limit	E	Ramp Time Factor = 2.5
C	Low Limit	F	Time

11.10 Supervisions

11.10.1 (ID 1431) Supervision #1 Item Selection

Location in the menu: **P3.8.1**

Use this parameter to select the supervision item. The output of the supervision function can be selected to the relay output.

11.10.2 (ID 1432) Supervision #1 Mode

Location in the menu: **P3.8.2**

Use this parameter to set the supervision mode.

When the 'Low limit' mode is selected, the output of the supervision function is active when the signal is below the supervision limit.

When the 'High limit' mode is selected, the output of the supervision function is active when the signal is above the supervision limit.

11.10.3 (ID 1433) Supervision #1 Limit

Location in the menu: **P3.8.3**

Use this parameter to set the supervision limit for the selected item. The unit shows automatically.

11.10.4 (ID 1434) Supervision #1 Limit Hysteresis

Location in the menu: **P3.8.4**

Use this parameter to set the supervision limit hysteresis for the selected item. The unit shows automatically.

11.10.5 (ID 1435) Supervision #2 Item Selection

Location in the menu: **P3.8.5**

Use this parameter to select the supervision item. The output of the supervision function can be selected to the relay output.

11.10.6 (ID 1436) Supervision #2 Mode

Location in the menu: **P3.8.6**

Use this parameter to set the supervision mode.

11.10.7 (ID 1437) Supervision #2 Limit

Location in the menu: **P3.8.7**

Use this parameter to set the supervision limit for the selected item. The unit shows automatically.

11.10.8 (ID 1438) Supervision #2 Limit Hysteresis

Location in the menu: **P3.8.8**

Use this parameter to set the supervision limit hysteresis for the selected item. The unit shows automatically.

11.11 Protections

11.11.1 General

11.11.1.1 (ID 701) Response to External Fault

Location in the menu: **P3.9.1.2**

Use this parameter to select the response of the drive to an 'External Fault'.

If a fault occurs, the drive can show a notification of it on the display of the drive. An external fault is activated with a digital input signal. The default digital input is DI3. It is also possible to program the response data into a relay output.

11.11.1.2 (ID 730) Input Phase Fault

Location in the menu: **P3.9.1.3**

Use this parameter to select the supply phase configuration of the drive.

If the 1-phase supply is used, the value of this parameter must be set to 1-phase support.

11.11.1.3 (ID 727) Undervoltage Fault

Location in the menu: **P3.9.1.4**

Use this parameter to select if undervoltage faults are saved to the fault history or not.

11.11.1.4 (ID 702) Response to Output Phase Fault

Location in the menu: **P3.9.1.5**

Use this parameter to select the response of the drive to an 'Output Phase' fault. If the measurement of the motor current detects that there is no current in 1 motor phase, an output phase fault occurs. See P3.9.1.2.

11.11.1.5 (ID 733) Response to Fieldbus Communication Fault

Location in the menu: **P3.9.1.6**

Use this parameter to select the response of the drive to a fieldbus communication fault. If the data connection between the master and the fieldbus board is defective, a fieldbus fault occurs.

11.11.1.6 (ID 734) Slot Communication Fault

Location in the menu: **P3.9.1.7**

Use this parameter to select the response of the drive to a 'Slot Communication' fault. If the drive detects a defective option board, a slot communication fault occurs. See P3.9.1.2.

11.11.1.7 (ID 732) Thermistor Fault

Location in the menu: **P3.9.1.8**

Use this parameter to select the response of the drive to a 'Thermistor' fault. If the thermistor detects too high temperature, a thermistor fault occurs. See P3.9.1.2.

11.11.1.8 (ID 748) PID Soft Fill Fault

Location in the menu: **P3.9.1.9**

Use this parameter to select the response of the drive to a 'PID Soft Fill' fault. If the PIDFeedback value does not reach the set level within the time limit, a soft fill fault occurs. See P3.9.1.2.

11.11.1.9 (ID 749) Response to PID Supervision Fault

Location in the menu: **P3.9.1.10**

Use this parameter to select the response of the drive to a 'PID Supervision' fault. If the PID feedback value is not within the supervision limits for longer than the supervision delay, a PID supervision fault occurs. See P3.9.1.2.

11.11.1.10 (ID 757) Response to External PID Supervision Fault

Location in the menu: **P3.9.1.11**

Use this parameter to select the response of the drive to a 'PID Supervision' fault. If the PID feedback value is not within the supervision limits for longer than the supervision delay, a PID supervision fault occurs. See P3.9.1.2.

11.11.1.11 (ID 703) Earth Fault

Location in the menu: **P3.9.1.12**

Use this parameter to select the response of the drive to an 'Earth Fault'.

If the measurement of the current detects that the sum of the motor phase currents is not 0, an earth fault occurs. See P3.9.1.2.

N O T I C E

It is possible to configure this fault only in enclosure sizes MR7, MR8, MR9, MR10, MR11, and MR12, and also MR6 600 V & 690 V.

11.11.1.12 (ID 183) Preset Alarm Frequency

Location in the menu: **P3.9.1.13**

Use this parameter to set the frequency of the drive when a fault is active and the response to the fault is set to 'Alarm + Preset Frequency'.

11.11.1.13 (ID 775) Response to Safe Torque Off (STO) Fault

Location in the menu: **P3.9.1.14**

Use this parameter to select the response of the drive to a 'STO Fault'.

This parameter defines drive operation when Safe Torque Off (STO) function is activated (for example, emergency stop button has been pressed or some other STO operation has been activated). See P3.9.1.2.

11.11.1.14 (ID 751) Fieldbus Communication Fault Force Delay

Location in the menu: **P3.9.1.17**

Use this parameter to set a delay for temporarily forcing the fault-response for fieldbus timeout from 'No Action', 'Alarm' or 'Alarm Preset Frequency' to 'Fault'. When a timeout occurs, the drive responds as selected with P3.9.1.6 and keeps running. After the delay time set by this parameter, the drive goes to a fault-state and the stop running.

If this parameter is set to 0.00 s, the fault-type remains as it is set with P3.9.1.6 and the drive runs indefinitely. If the fault-type is Fault or Fault-Coast, this delay does not affect anything.

11.11.1.15 (ID 752) Fieldbus Communication Fault Clear Delay

Location in the menu: **P3.9.1.18**

Use this parameter to set a delay for accepting new fieldbus start, reverse, and reference commands after a timeout has been cleared. The delay starts running after the connection has been re-established, and not after clearing a fieldbus timeout fault.

11.11.1.16 (ID 753) Unbalance Current Alarm

Location in the menu: **P3.9.1.19**

Use this parameter to set the response of the drive to an Unbalance Current Alarm. Available options are 'No Action' and 'Alarm'. This option is only available for MR11 and MR12 enclosure sizes.

11.11.2 Motor Thermal Protections

The motor thermal protection prevents the motor from becoming too hot.

The AC drive can supply a current that is higher than the nominal current. The high current can be necessary to the load, and it must be used. In these conditions, there is a risk of a thermal overload. Low frequencies have a higher risk. At low frequencies, the cooling effect and the capacity of the motor decrease. If the motor has an external fan, the load reduction at low frequencies is small.

The motor thermal protection is based on calculations. The protection function uses the output current of the drive to know what is the load on the motor. If the control board is not energized, the calculations are reset.

To adjust the thermal protection of the motor, use the parameters from P3.9.2.1 to P3.9.2.5. It is possible to monitor the thermal status of the motor on the display of the control panel.

N O T I C E

If long motor cables (maximum 100 m) are used with small drives (≤ 1.5 kW), the motor current that the drive measures can be much higher than the actual motor current. It is because there are capacitive currents in the motor cable.

N O T I C E

OVERHEATING BECAUSE OF BLOCKED AIRFLOW

If the airflow to the motor is blocked, the function does not protect the motor, and the motor can become too hot. It can damage the motor.

- Make sure that the airflow to the motor is not blocked.

11.11.2.1 (ID 704) Motor Thermal Protection

Location in the menu: **P3.9.2.1**

Use this parameter to select the response of the drive to a 'Motor Overtemperature' fault. If the motor thermal protection function detects that the temperature of the motor is too high, a motor overtemperature fault occurs.

If possible, use a motor thermistor to protect the motor. Set the value of this parameter to 0.

11.11.2.2 (ID 705) Ambient Temperature

Location in the menu: **P3.9.2.2**

Use this parameter to set the ambient temperature where the motor is installed. The temperature value is given in Celsius or Fahrenheit degrees.

11.11.2.3 (ID 706) Zero Speed Cooling Factor

Location in the menu: **P3.9.2.3**

Use this parameter to set the cooling factor at 0 speed in relation to the point where the motor operates at nominal speed without external cooling.

The default value is set for conditions where there is no external fan. When using an external fan, the value can be set higher than without the fan, for example at 90%.

When changing parameter P3.1.1.4 (Motor Nominal Current), parameter P3.9.2.3 is automatically set to its default value.

Changing this parameter does not affect the maximum output current of the drive. Only parameter P3.1.3.1 *Motor Current Limit* can change the maximum output current.

The corner frequency for the thermal protection is 70% of the value of the parameter P3.1.1.2 Motor Nominal Frequency.

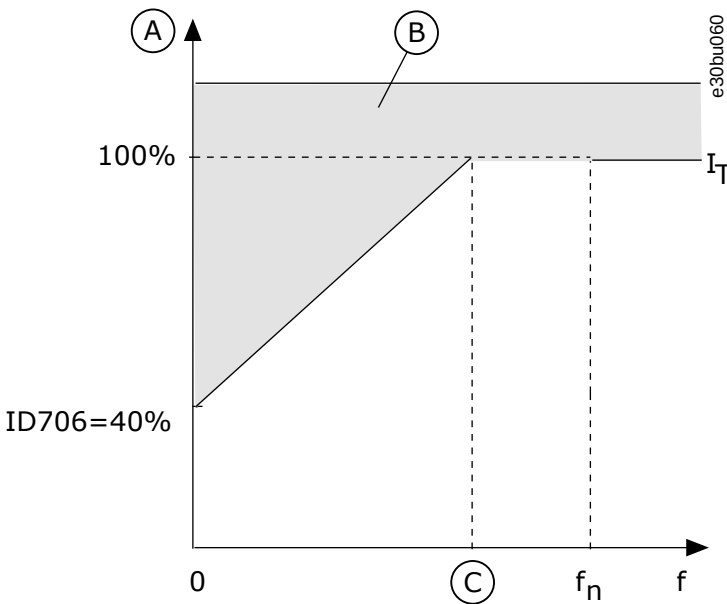


Illustration 65: The Motor Thermal Current I_T Curve

A	$P_{Cooling}$	C	Corner frequency
B	Overload area		

11.11.2.4 (ID 707) Motor Thermal Time Constant

Location in the menu: **P3.9.2.4**

Use this parameter to set the motor thermal time constant.

The time constant is the time within which the calculated thermal stage has reached 63% of its final value. The final thermal stage equals to running the motor continuously with nominal load at nominal speed. The length of the time constant is in relation with the dimension of the motor. The bigger the motor, the longer the time constant.

In different motors, the motor thermal time constant is different. It also changes between different motor manufacturers. The default value of the parameter changes from dimension to dimension.

The t_6 -time is the time in seconds that the motor can safely operate at 6 times the rated current. It is possible that the motor manufacturer gives the data with the motor. Use this information to set the time constant parameter. Usually, the motor thermal time constant in minutes is $2 \cdot t_6$. When the drive is in the *STOP* state, the time constant is internally increased to 3 times the set parameter value, because the cooling operates based on convection.

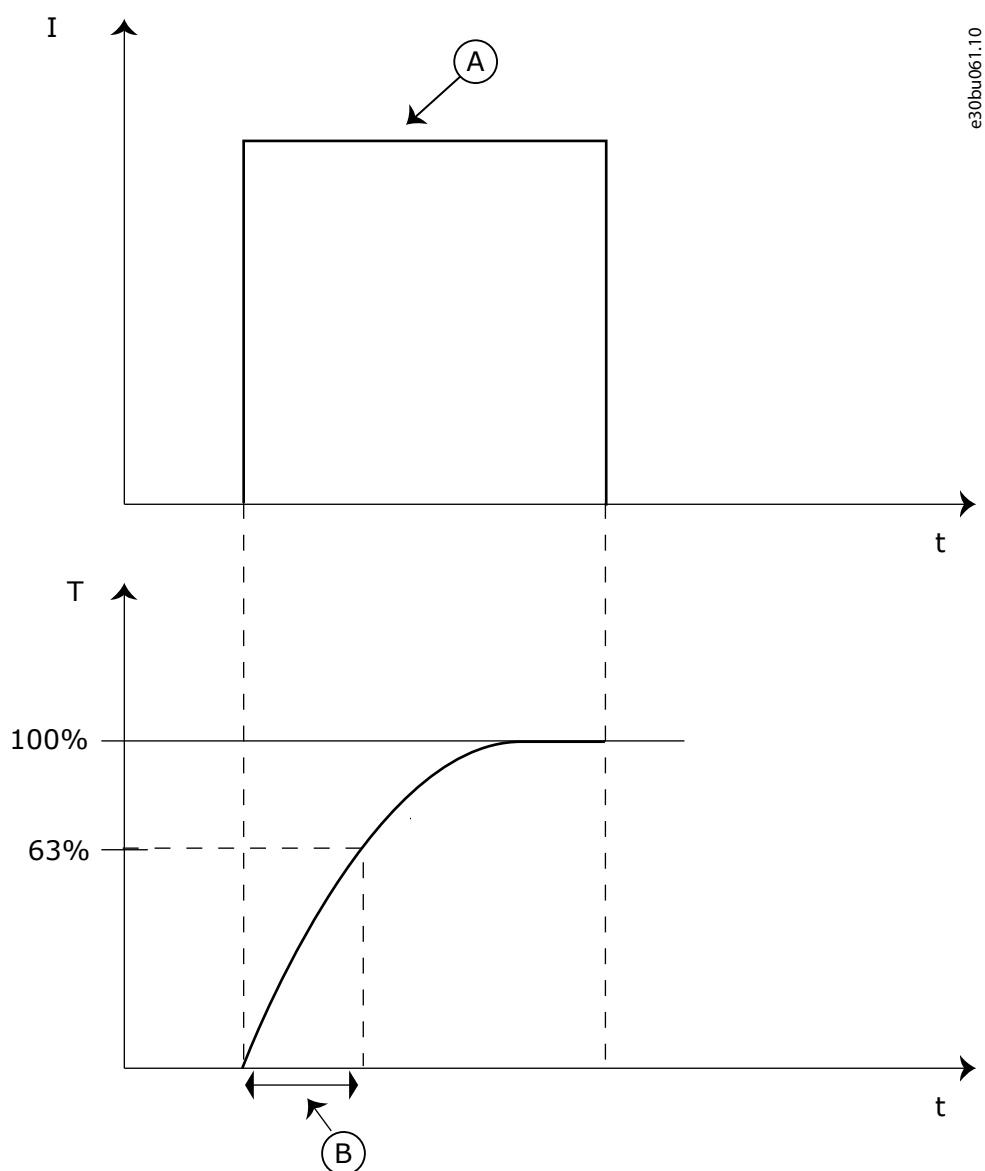


Illustration 66: The Motor Thermal Time Constant

A	Current
B	Motor thermal time constant

11.11.2.5 (ID 708) Motor Thermal Loadability

Location in the menu: **P3.9.2.5**

Use this parameter to set the thermal loadability of the motor. For example, if the value is set to 130%, the motor goes to the nominal temperature with 130% of the motor nominal current.

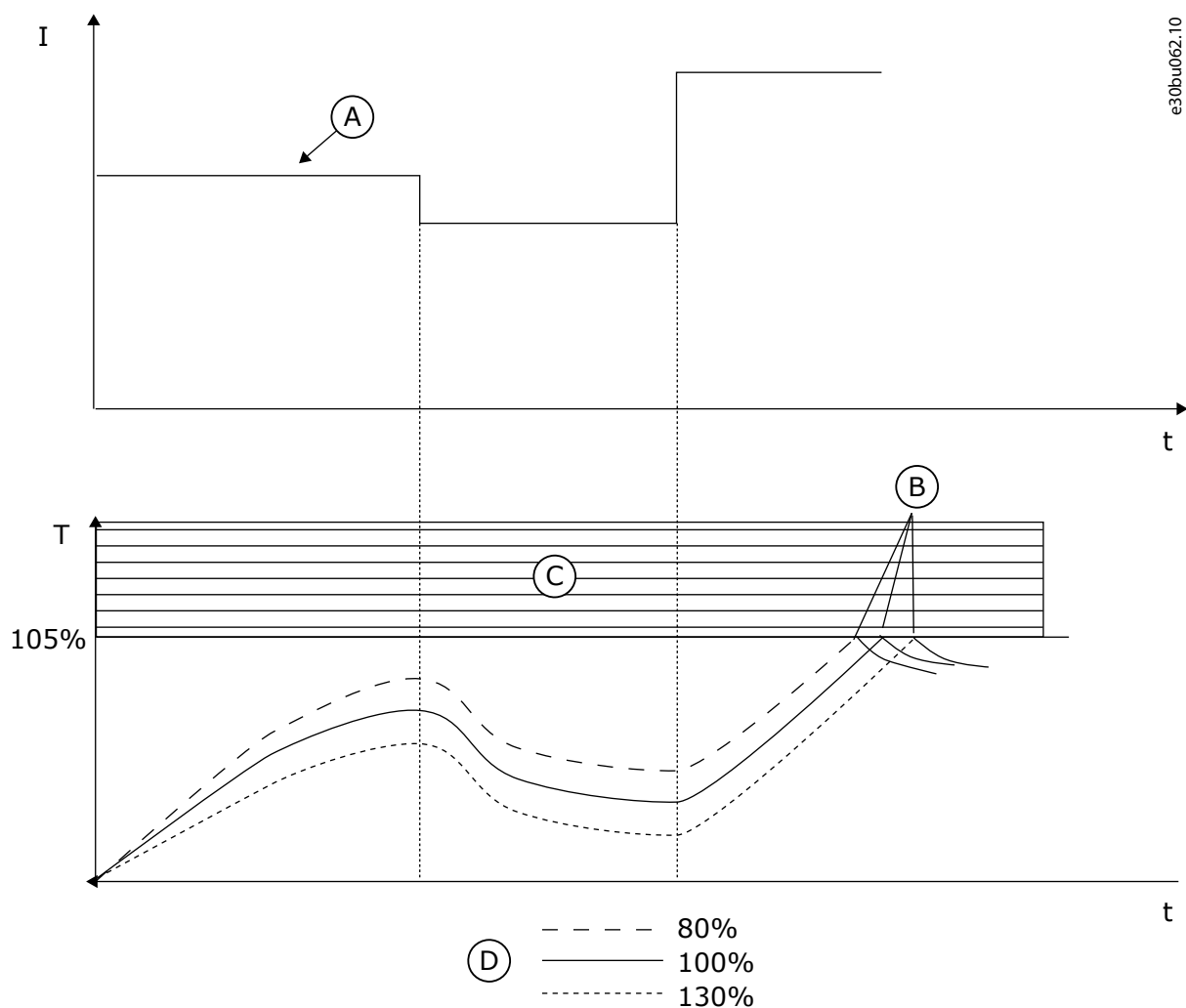


Illustration 67: The Calculation of the Motor Temperature

A	Current	C	Trip area
B	Fault/Alarm	D	Loadability

11.11.3 Motor Stall Protection

The motor stall protection function gives protection to the motor against short overloads. An overload can be caused, for example, by a stalled shaft. It is possible to set the reaction time of the stall protection shorter than that of the motor thermal protection.

The stall status of the motor is specified with parameters P3.9.3.2 Stall Current and P3.9.3.4 Stall Frequency Limit. If the current is higher than the limit, and the output frequency is lower than the limit, the motor is in a stall status.

The stall protection is a type of overcurrent protection.

NOTICE

If long motor cables (maximum 100 m) are used with small drives (≤ 1.5 kW), the motor current that the drive measures can be much higher than the actual motor current. It is because there are capacitive currents in the motor cable.

11.11.3.1 (ID 709) Motor Stall Fault

Location in the menu: **P3.9.3.1**

Use this parameter to select the response of the drive to a 'Motor Stall' fault. If the stall protection detects that the shaft of the motor is stalled, a motor stall fault occurs.

11.11.3.2 (ID 710) Stall Current

Location in the menu: **P3.9.3.2**

Use this parameter to set the limit above which the current of the motor must stay for a stall stage to occur.

It is possible to set the value of this parameter between 0.0 and $2 \cdot I_L$. For a stall status to occur, the current must be higher than this limit. If parameter P3.1.3.1 Motor Current Limit changes, this parameter is automatically calculated to 90% of the current limit.

The value of the Stall Current must be below the motor current limit.

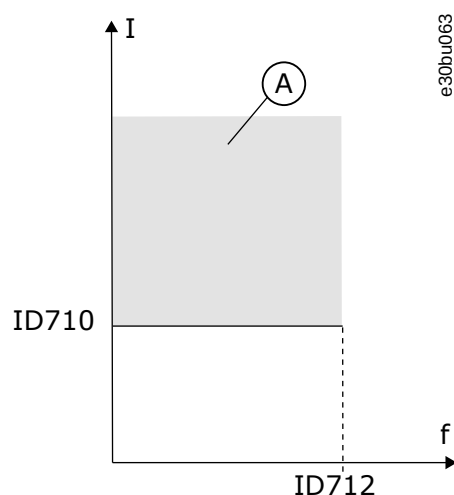


Illustration 68: The Stall Characteristics Settings

A	Stall area
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11.11.3.3 (ID 711) Stall Time Limit

Location in the menu: **P3.9.3.3**

Use this parameter to set the maximum time for a stall stage.

This parameter is the maximum time for the stall stage to be active before a motor stall fault occurs. The value of this parameter can be set between 1.0 s and 120.0 s. An internal counter counts the stall time. If the stall time counter value goes above this limit, the protection causes the drive to trip.

11.11.3.4 (ID 712) Stall Frequency Limit

Location in the menu: **P3.9.3.4**

Use this parameter to set the limit below which the output frequency of the drive must stay for a stall stage to occur.

For a stall state to occur, the output frequency must be below this limit for a certain time.

11.11.4 Underload (Dry Pump) Protection

The motor underload protection makes sure that there is a load on the motor when the drive operates. If the motor loses the load, a problem can occur in the process. For example, a belt can break or a pump become dry.

The motor underload protection can be adjusted with parameters P3.9.4.2 (Underload Protection: Field Weakening Area Load) and P3.9.4.3 (Underload Protection: Zero Frequency Load). The underload curve is a squared curve between the zero frequency and the field weakening point. The protection is not active below 5 Hz. The underload time counter does not operate below 5 Hz.

The values of the underload protection parameters are set in percentage of the nominal torque of the motor. To find the scaling ratio for the internal torque value, use the data in the nameplate data of the motor, the motor nominal current, and the nominal current of the drive I_H . If another current than the nominal motor current is used, the precision of the calculation decreases.

NOTICE

If long motor cables (maximum 100 m) are used with small drives (≤ 1.5 kW), the motor current that the drive measures can be much higher than the actual motor current. It is because there are capacitive currents in the motor cable.

11.11.4.1 (ID 713) Underload Fault

Location in the menu: **P3.9.4.1**

Use this parameter to select the response of the drive to an 'Underload' fault. If the underload protection function detects that there is not a sufficient load on the motor, an underload fault occurs.

11.11.4.2 (ID 714) Underload Protection: Field Weakening Area Load

Location in the menu: **P3.9.4.2**

Use this parameter to set the minimum torque that the motor needs when the output frequency of the drive is higher than the frequency of the weakening point. It is possible to set the value of this parameter between 10.0% and 150.0% $\times T_{nMotor}$. This value is the limit for the minimum torque when the output frequency is above the field weakening point.

Changing parameter P3.1.1.4 (Motor Nominal Current) restores this parameter automatically to its default value. See [11.11.4 Underload \(Dry Pump\) Protection](#).

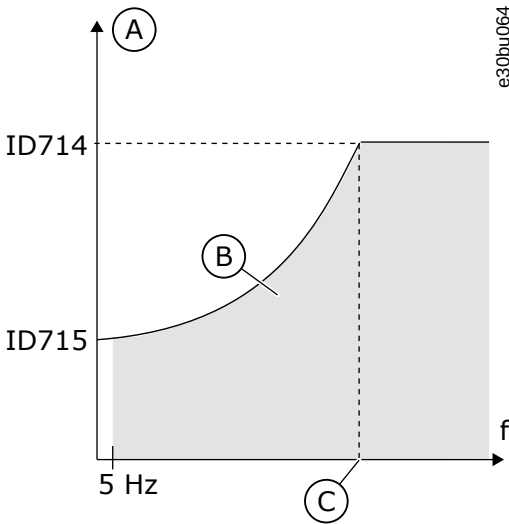


Illustration 69: Setting of the Minimum Load

A	Torque	C	Field weakening point
B	Underload area		

11.11.4.3 (ID 715) Underload Protection: Zero Frequency Load

Location in the menu: **P3.9.4.3**

Use this parameter to set the minimum torque that the motor needs when the output frequency of the drive is 0.

Changing the value of parameter P3.1.1.4 restores this parameter automatically to the default value.

11.11.4.4 (ID 716) Underload Protection: Time Limit

Location in the menu: **P3.9.4.4**

Use this parameter to set the maximum time for an underload state. It is the maximum time for the underload state to be active before an underload fault occurs.

It is possible to set the time limit between 2.0 s and 600.0 s.

An internal counter counts the underload time. If the value of the counter goes above this limit, the protection causes the drive to trip. The drive trips as is set in parameter P3.9.4.1 Underload Fault. If the drive stops, the underload counter goes back to 0.

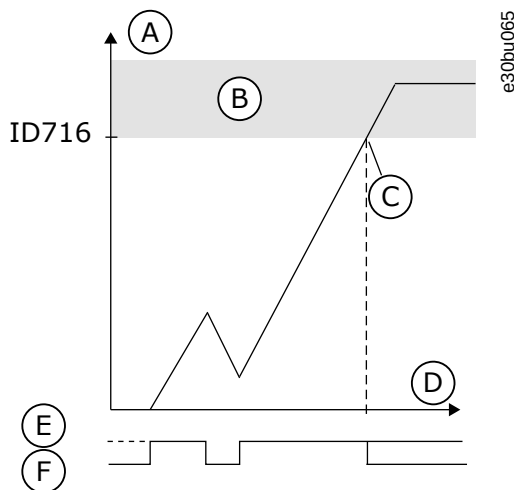


Illustration 70: The Underload Time Counter Function

A	Underload time counter	D	Time
B	Trip area	E	Underload
C	Trip/warning ID 713	F	No underload

11.11.5 Quick Stop

11.11.5.1 (ID 1276) Quick Stop Mode

Location in the menu: **P3.9.5.1**

Use this parameter to select how the drive stops when the quick stop command is given from DI or Fieldbus.

11.11.5.2 (ID 1213) Quick Stop Activation

Location in the menu: **P3.9.5.2**

Use this parameter to select the digital input signal that activates a Quick Stop function. The Quick Stop function stops the drive regardless of the control place or the state of the control signals.

11.11.5.3 (ID 1256) Quick Stop Deceleration Time

Location in the menu: **P3.9.5.3**

Use this parameter to set the time that is necessary for the output frequency to decrease from maximum frequency to 0 when a quick stop command is given. The value of this parameter is applied only when the quick stop mode parameter is set to 'Quick Stop Deceleration time'.

11.11.5.4 (ID 744) Response to Quick Stop Fault

Location in the menu: **P3.9.5.4**

Use this parameter to select the response of the drive to a 'Quick Stop' fault. If the quick stop command is given from DI or Fieldbus, a quick stop fault occurs.

With the quick stop function, it is possible to stop the drive in an unusual procedure from I/O or Fieldbus in unusual conditions. When the quick stop function is active, it is possible to make the drive decelerate and stop. It is possible to program an alarm or fault to put a mark in the fault history that there was a request for a quick stop.

⚠ CAUTION ⚠

Do not use the quick stop function as an emergency stop. An emergency stop must stop the power supply to the motor. The quick stop function does not do this.

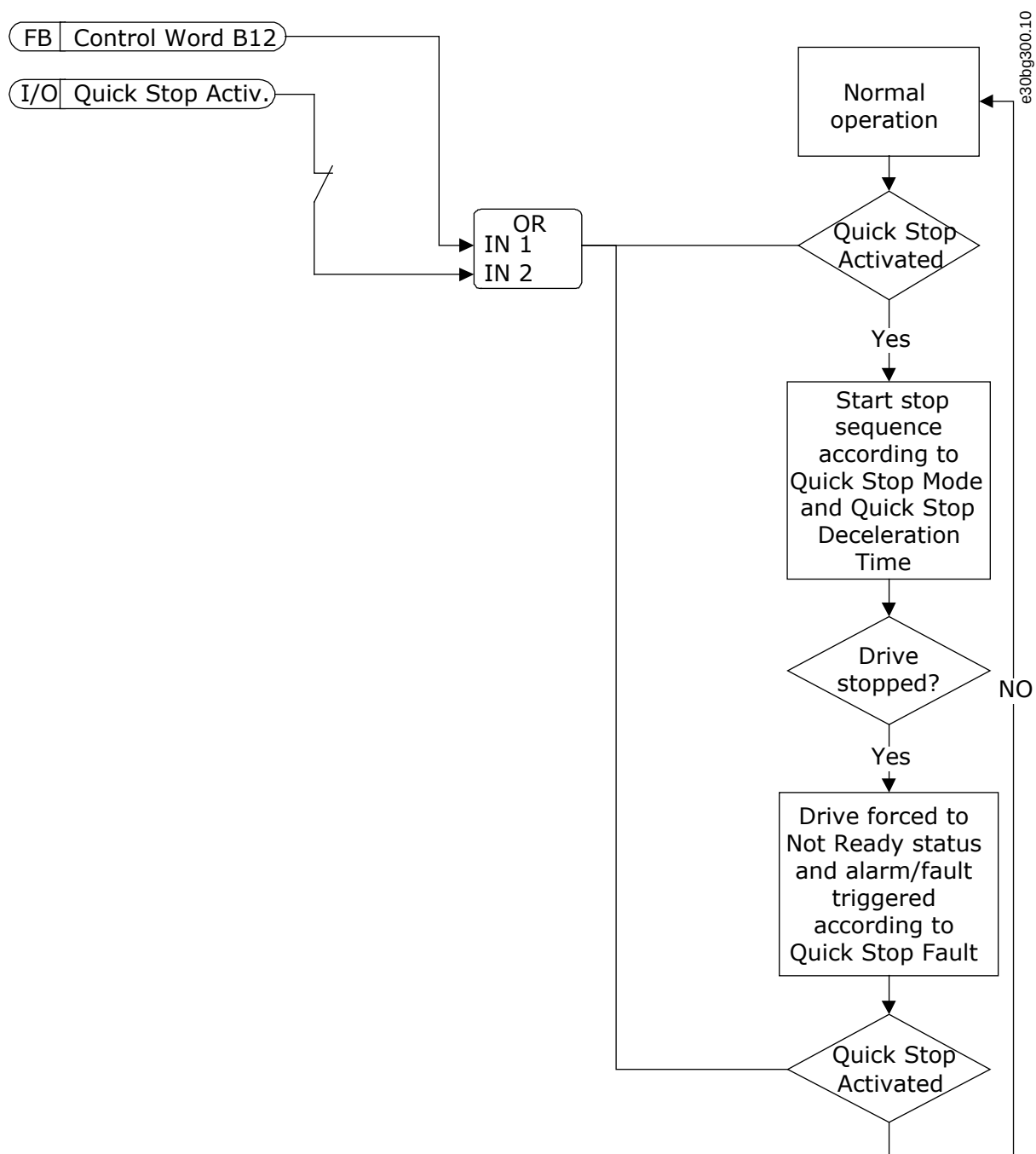


Illustration 71: The Quick Stop Logic

11.11.6 AI Low Protection

11.11.6.1 (ID 767) Analog Input Low Protection

Location in the menu: **P3.9.8.1**

Use this parameter to select when the AI Low supervision is enabled. For example, the AI Low supervision can be enabled only when the drive is in run state.

Use the AI Low Protection to find failures in the analog input signals. This function gives protection only to the analog inputs that are used as frequency reference, torque reference, or in the PID/ExtPID controllers.

It is possible to have the protection on when the drive is in the RUN status, or in the RUN and STOP statuses.

Selection number	Selection name	Description
0	Protection disabled	
1	Protection enabled in RUN status	The protection is enabled only when the drive is in the RUN status.
2	Protection enabled in RUN and STOP status	The protection is enabled in the 2 statuses, RUN, and STOP.

11.11.6.2 (ID 700) Analog Input Low Fault

Location in the menu: **P3.9.8.2**

Use this parameter to select the response of the drive to an 'AI Low' fault. If the analog input signal becomes less than 50% of the minimum signal for 500 ms, an AI Low fault occurs.

If AI Low Protection is enabled with parameter P3.9.8.1, this parameter gives a response for the fault code 50 (Fault ID 1050).

The AI low protection function monitors the signal level of the analog inputs 1–6. If the analog input signal becomes less than 50% of the minimum signal for 500 ms, an AI Low fault, or alarm shows.

NOTICE

It is possible to use the value Alarm + previous frequency only when using analog input 1 or analog input 2 as frequency reference.

Selection number	Selection name	Description
0	No Action	AI Low Protection is not used.
1	Alarm	
2	Alarm, preset frequency	The frequency reference is set as in P3.9.1.13 Preset Alarm Frequency.
3	Alarm, previous frequency	The last valid frequency is kept as frequency reference.
4	Fault	The drive stops as is set in P3.2.5 Stop Mode.
5	Fault, coasting	The drive stops by coasting.

11.12 Automatic Reset

11.12.1 (ID 731) Automatic Reset

Location in the menu: **P3.10.1**

Use this parameter to enable the Automatic reset function. To select the faults that are reset automatically, enter the value 0 or 1 to parameters from P3.10.6 to P3.10.13.

NOTICE

The automatic reset function is available only for some fault types.

11.12.2 (ID 719) Restart Function

Location in the menu: **P3.10.2**

Use this parameter to select the start mode for the Automatic reset function.

11.12.3 (ID 717) Wait Time

Location in the menu: **P3.10.3**

Use this parameter to set the wait time before the first reset is done.

11.12.4 (ID 718) Trial Time

Location in the menu: **P3.10.4**

Use this parameter to set the trial time for the automatic reset function. During the trial time, the automatic reset function tries to reset the faults that occur.

The time count starts from the first automatic reset. The next fault starts the trial time count again.

11.12.5 (ID 759) Number of Trials

Location in the menu: **P3.10.5**

Use this parameter to set the total number of auto reset trials. If the number of trials during the trial time is more than the value of this parameter, a permanent fault shows. If not, the fault goes out of view after the trial time is completed. The fault type does not affect the maximum number of trials.

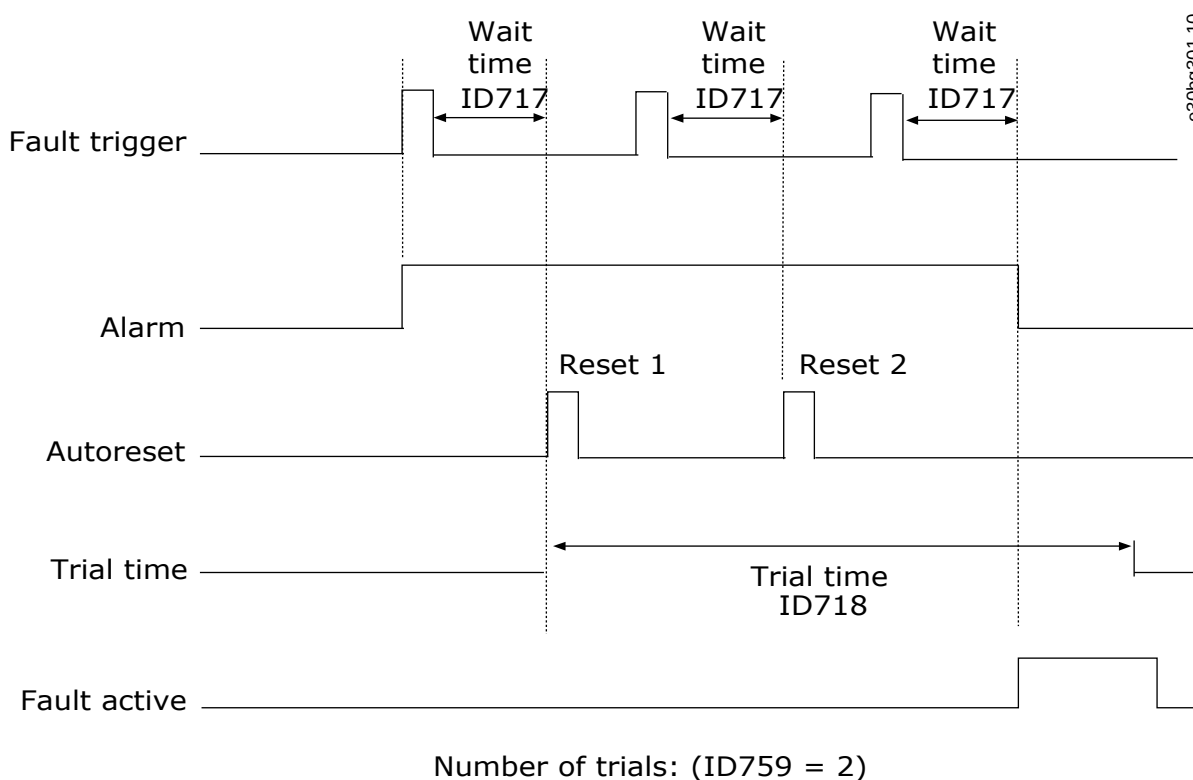


Illustration 72: The Automatic Reset Function

11.12.6 (ID 720) Autoreset: Undervoltage

Location in the menu: **P3.10.6**

Use this parameter to enable the automatic reset after an undervoltage fault.

11.12.7 (ID 721) Autoreset: Overvoltage

Location in the menu: **P3.10.7**

Use this parameter to enable the automatic reset after an overvoltage fault.

11.12.8 (ID 722) Autoreset: Overcurrent

Location in the menu: **P3.10.8**

Use this parameter to enable the automatic reset after an overcurrent fault.

11.12.9 (ID 723) Autoreset: AI Low

Location in the menu: **P3.10.9**

Use this parameter to enable the automatic reset after a fault caused by low AI signal.

11.12.10 (ID 724) Autoreset: Unit Overtemperature

Location in the menu: **P3.10.10**

Use this parameter to enable the automatic reset after a fault caused by unit overtemperature.

11.12.11 (ID 725) Autoreset: Motor Overtemperature

Location in the menu: **P3.10.11**

Use this parameter to enable the automatic reset after a fault caused by motor overtemperature.

11.12.12 (ID 726) Autoreset: External Fault

Location in the menu: **P3.10.12**

Use this parameter to enable the automatic reset after an external fault.

11.12.13 (ID 738) Autoreset: Underload Fault

Location in the menu: **P3.10.13**

Use this parameter to enable the automatic reset after an underload fault.

11.12.14 (ID 776) Autoreset: PID Supervision Fault

Location in the menu: **P3.10.14**

Use this parameter to enable the automatic reset after a PID supervision fault.

11.12.15 (ID 777) Autoreset: EXT PID Supervision Fault

Location in the menu: **P3.10.15**

Use this parameter to enable the automatic reset after an external PID supervision fault.

11.12.16 (ID 1569) Fault Simulation

Location in the menu: **P3.10.16**

With this parameter it is possible to simulate the different faults without them being actually active. For example simulating an over current situation. In the drive interface point of view the behavior is identical to actual fault situations.

Selection	Name	Description
B00	1 ₁₀	Over current fault (F1)
B01	2 ₁₀	Over voltage fault (F2)
B02	4 ₁₀	Under voltage fault (F9)
B03	8 ₁₀	Output phase supervision fault (F11)
B04	16 ₁₀	Earth voltage fault (F3)
B05	32 ₁₀	System fault (F8)
B06	64 ₁₀	Over temperature warning (W14)
B07	128 ₁₀	Over temperature fault (F14) ⁽¹⁾
B08	256 ₁₀	Reserved
B09	512 ₁₀	Reserved
B10	1024 ₁₀	Fan cooling fault (F32)

¹ The warning bit needs to be high also for this fault simulation to work.

11.13 Application Settings

11.13.1 (ID 1806) Password

Location in the menu: **P3.11.1**

Use this parameter to set the administrator password.

11.13.2 (ID 1197) C/F Selection

Location in the menu: **P3.11.2**

Use this parameter to set the temperature measuring unit. The system shows all the temperature-related parameters and monitoring values in the set unit.

11.13.3 (ID 1198) kW/HP Selection

Location in the menu: **P3.11.3**

Use this parameter to set the power measuring unit. The system shows all the power-related parameters and monitoring values in the set unit.

11.13.4 (ID 1196) Multimonitor View

Location in the menu: **P3.11.4**

Use this parameter to set the division of the display of the control panel into sections in the multimonitor view.

11.13.5 (ID 1195) [FUNCT] Button Configuration

Location in the menu: **P3.11.5**

Use this parameter to set the values of the [FUNCT] button. The values set with this parameter are available when pushing the [FUNCT] button on the keypad.

11.13.6 (ID 1807) Set Parameter Lock

Location in the menu: **P3.11.6**

Use this parameter to set a password for edit-locking all parameters. Parameters are locked if value is set greater than zero. To disable the lock, open the lock with a correct key, and set the lock-password back to zero.

NOTE! After setting this parameter to larger than zero (0), the lock is immediately active. Setting a lock also locks and hide this parameter itself. Control place cannot be changed over to Local, when the lock is in place.

11.13.7 (ID 1808) Parameter Lock Key

Location in the menu: **P3.11.7**

Use this parameter to enter a password for opening the parameter edit-lock. The lock is opened if the values of the Key- and Lock-passwords are equal.

11.14 Timer Functions

11.14.1 Introduction to Timer Functions

The timer functions make it possible for the internal RTC (real-time clock) to control functions. All the functions that can be controlled with a digital input, can also be controlled with the RTC, with time channels 1–3. It is not necessary to have an external PLC to control a digital input. It is possible program the closed and opened intervals of the input internally.

To get the best results of the timer functions, install a battery, and make the settings of the real-time clock carefully in the Start-up wizard. The battery is available as an option.

NOTICE

It is not recommended to use the timer functions without an auxiliary battery. The time and date settings of the drive are reset at each power down, if there is no battery for the RTC.

11.14.2 Time Channels, Intervals, and Timers

Time Channels

It is possible to assign the output of the interval and/or timer functions to time channels 1–3. It is possible to use the time channels to control on/off type functions, for example relay outputs or digital inputs. To configure the on/off logic of the time channels, assign intervals and/or timers to them. A time channel can be controlled by many different intervals or timers.

Assigning intervals and timers to time channels is flexible. Every interval and timer has a parameter with which to assign them to a time channel.

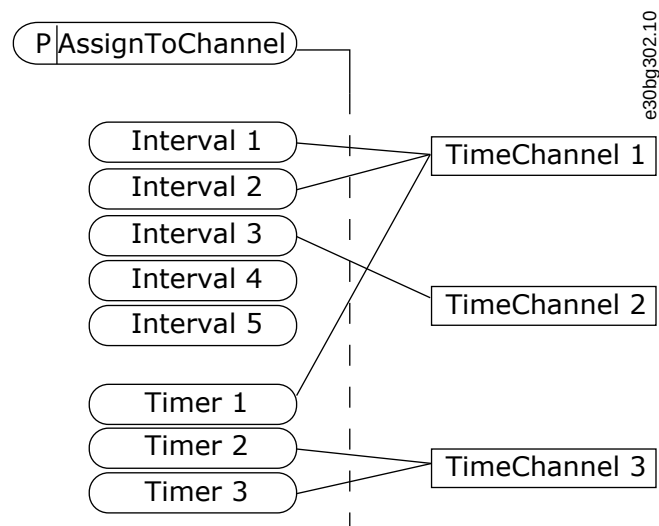


Illustration 73: Assigning Intervals and Timers to time Channels

Intervals

Use parameters to give each interval an ON Time and OFF Time. It is the daily active time of the interval during the days set with parameters From Day and To Day. For example, with the parameter settings below, the interval is active from 7 am to 9 am from Monday to Friday. The time channel is like a digital input, but virtual.

- ON Time: 07:00:00
- OFF Time: 09:00:00
- From Day: Monday
- To Day: Friday

Timers

Use the timers to set a time channel as active for a period with a command from a digital input or a time channel.

In the following figure, the activation signal comes from a digital input or a virtual digital input, like a time channel. The timer counts down from the falling edge.

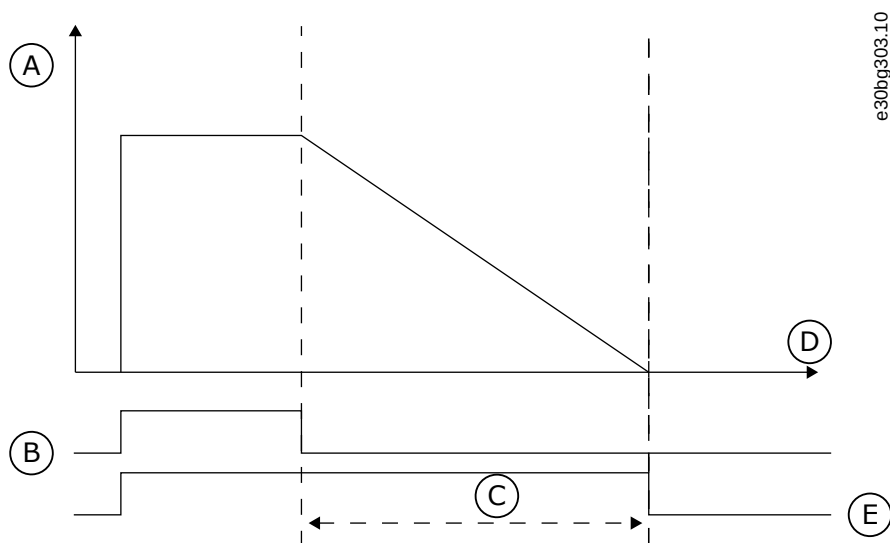


Illustration 74: The Activation Signal from Digital Input or Virtual Digital Input

A	Remaining time	D	Time
B	Activation	E	OUT
C	Duration		

The following parameters set the timer active when the digital input 1 on the slot A is closed. They will also keep the timer active for 30 s after it is opened.

- Duration: 30 s
- Timer: DigIn SlotA.1

It is possible use a duration of 0 s to override a time channel that is activated from a digital input. There is no off delay after the falling edge.

Example

Problem: The AC drive is in a warehouse and controls air conditioning. It must operate between 7 am and 5 pm on weekdays and between 9 am and 1 pm on weekends. It is also necessary for the drive to operate outside these hours, if there are personnel in the building. The drive must continue to operate 30 minutes after the personnel has left.

Solution: Set 2 intervals, 1 for weekdays and 1 for weekends. A timer is also necessary to activate the process outside the set hours. See the following configuration.

Interval 1

- P3.12.1.1: ON Time: 07:00:00
- P3.12.1.2: OFF Time: 17:00:00
- P3.12.1.3: Days: Monday, Tuesday, Wednesday, Thursday, Friday
- P3.12.1.4: Assign to channel: Time channel 1

Interval 2

- P3.12.2.1: ON Time: 09:00:00
- P3.12.2.2: OFF Time: 13:00:00
- P3.12.2.3: Days: Saturday, Sunday
- P3.12.2.4: Assign to channel: Time channel 1

Timer 1

- P3.12.6.1: Duration: 1800 s (30 min)
- P3.12.6.2: Timer 1: DigIn SlotA.1 (The parameter is in the digital inputs menu.)
- P3.12.6.3: Assign to channel: Time channel 1
- P3.5.1.1: Control signal 1 A: Time Channel 1 for the I/O Run command

In the following figure, Time Channel 1 is used as the control signal for the start command instead of a digital input.

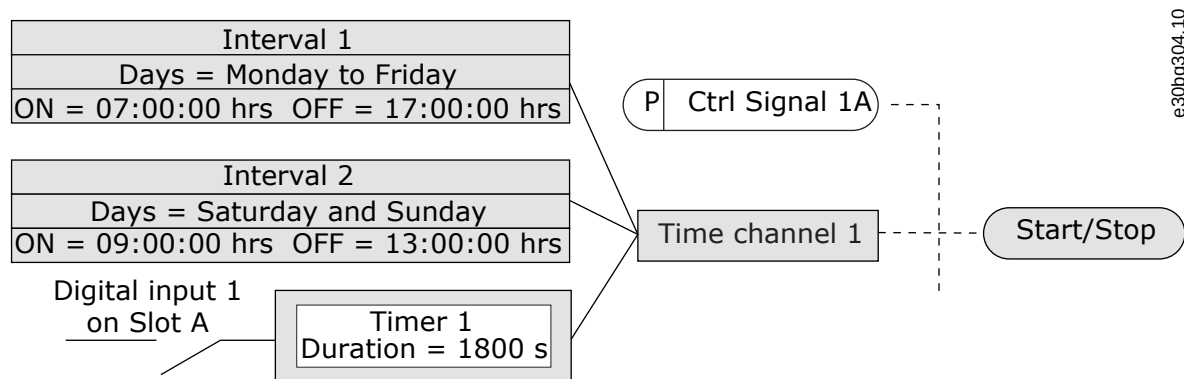
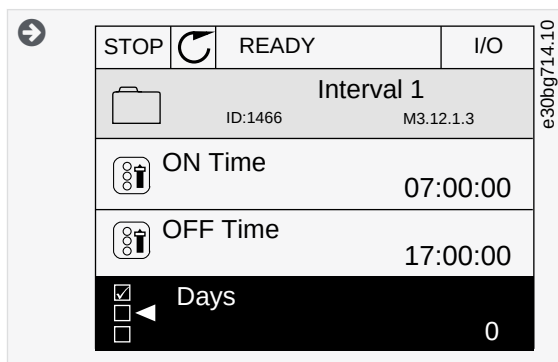


Illustration 75: Time Channel 1 as the Control Signal for the Start Command

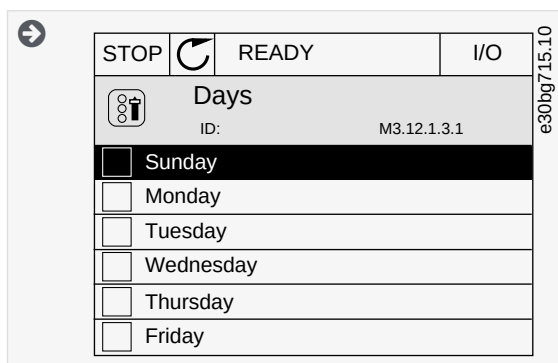
11.14.3 Editing an Interval

Procedure

1. Activate the interval to edit.
2. Go into parameter Days (ID 1466).



3. Go into the Edit mode.
4. Select the necessary weekdays in the checkbox selection.



11.14.4 (ID 1464) ON Time

Location in the menu: **P3.12.1.1**

Use this parameter to set the time of day when the output of the interval function is activated.

11.14.5 (ID 1465) OFF Time

Location in the menu: **P3.12.1.2**

Use this parameter to set the time of day when the output of the interval function is deactivated.

11.14.6 (ID 1466) Days

Location in the menu: **P3.12.1.3**

Use this parameter to select the days of the week when the interval function is enabled.

11.14.7 (ID 1468) Assign to Channel

Location in the menu: **P3.12.1.4**

Use this parameter to select the time channel where the output of the interval function is assigned. It is possible to use the time channels to control the on/off type functions, for example relay outputs or any functions that can be controlled by a DI signal.

11.14.8 (ID 1489) Duration

Location in the menu: **P3.12.6.1**

Use this parameter to set the duration that the timer runs when the activation signal is removed (Off-delay).

11.14.9 (ID 447) Timer 1

Location in the menu: **P3.12.6.2**

Use this parameter to select the digital input signal that starts the timer. The output of the timer is activated when this signal is activated. The timer starts to count when this signal is deactivated (falling edge). The output is deactivated when the time that is set with the duration parameter has elapsed. The rising edge starts Timer 1 that is programmed in Group 3.12.

11.14.10 (ID 1490) Assign to Channel

Location in the menu: **P3.12.6.3**

Use this parameter to select the time channel where the output of the timer function is assigned. It is possible to use the time channels to control the on/off type functions, for example relay outputs or any functions that can be controlled by a DI signal.

11.15 PID Controller

11.15.1 Basic Settings

11.15.1.1 (ID 118) PID Gain

Location in the menu: **P3.13.1.1**

Use this parameter to adjust the gain of the PID controller. If this parameter is set to 100%, a change of 10% in the error value causes the controller output to change by 10%.

11.15.1.2 (ID 119) PID Integration Time

Location in the menu: **P3.13.1.2**

Use this parameter to adjust the integration time of the PID controller. If this parameter is set to 1.00 s, a change of 10% in the error value causes the controller output to change by 10.00%/s.

11.15.1.3 (ID 132) PID Derivation Time

Location in the menu: **P3.13.1.3**

Use this parameter to adjust the derivation time of the PID controller. If this parameter is set to 1.00 s, a change of 10% in the error value during 1.00 s causes the controller output to change by 10.00%.

11.15.1.4 (ID 1036) Process Unit Selection

Location in the menu: **P3.13.1.4**

Use this parameter to select the unit for the feedback and the setpoint signals of the PID controller. Select the unit for the actual value.

11.15.1.5 (ID 1033) Process Unit MIN

Location in the menu: **P3.13.1.5**

Use this parameter to set the minimum value of the PID feedback signal. For example, an analog signal of 4...20 mA corresponds to the pressure of 0...10 bar. The value in process units at a 0% feedback or setpoint. This scaling is done for monitoring purposes only. The PID controller still uses the percentage internally for feedbacks and setpoints.

11.15.1.6 (ID 1034) Process Unit MAX

Location in the menu: **P3.13.1.6**

Use this parameter to set the maximum value of the PID feedback signal. For example, an analog signal of 4...20 mA corresponds to the pressure of 0...10 bar. The value in process units at a 0% feedback or setpoint. This scaling is done for monitoring purposes only. The PID controller still uses the percentage internally for feedbacks and setpoints.

11.15.1.7 (ID 1035) Process Unit Decimals

Location in the menu: **P3.13.1.7**

Use this parameter to set the number of decimals for the process unit values. For example, an analog signal of 4...20 mA corresponds to the pressure of 0...10 bar. The value in process units at a 0% feedback or setpoint. This scaling is done for monitoring purposes only. The PID controller still uses the percentage internally for feedbacks and setpoints.

11.15.1.8 (ID 340) Error Inversion

Location in the menu: **P3.13.1.8**

Use this parameter to invert the error value of the PID controller.

11.15.1.9 (ID 1056) Dead Band

Location in the menu: **P3.13.1.9**

Use this parameter to set the dead band area around the PID setpoint value. The value of this parameter is given in the selected process unit. The output of the PID controller is locked if the feedback value stays in the dead band area for the set time.

11.15.1.10 (ID 1057) Dead Band Delay

Location in the menu: **P3.13.1.10**

Use this parameter to set the time that the feedback value must stay in the dead band area before the output of the PID controller is locked. If the actual value stays in the dead band area for a time set in Dead Band Delay, the PID controller output is locked. This function prevents wear and unwanted movements of the actuators, for example valves.

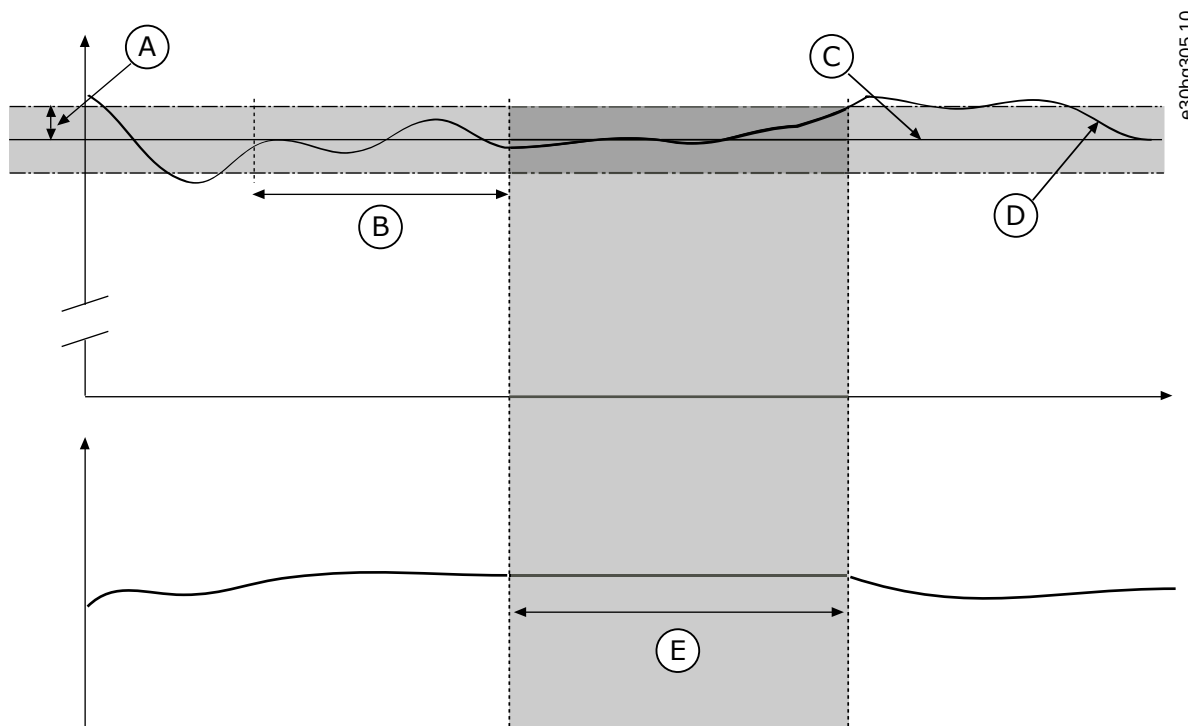


Illustration 76: The Dead Band Function

A	Dead band (ID1056)	D	Actual value
B	Dead band delay (ID1057)	E	Output locked
C	Reference		

11.15.2 Setpoints

11.15.2.1 (ID 167) Keypad Setpoint 1

Location in the menu: **P3.13.2.1**

Use this parameter to set the setpoint value of the PID controller when the setpoint source is 'Keypad SP'. The value of this parameter is given in the selected process unit.

11.15.2.2 (ID 168) Keypad Setpoint 2

Location in the menu: **P3.13.2.2**

Use this parameter to set the setpoint value of the PID controller when the setpoint source is 'Keypad SP'. The value of this parameter is given in the selected process unit.

11.15.2.3 (ID 1068) Setpoint Ramp Time

Location in the menu: **P3.13.2.3**

Use this parameter to set the rising and falling ramp times for the setpoint changes. Ramp time is the time that is necessary for the setpoint value to change from minimum to maximum. If the value of this parameter is set to 0, no ramps are used.

11.15.2.4 (ID 1046) PID Setpoint Boost Activation

Location in the menu: **P3.13.2.4**

Use this parameter to select the digital input signal that activates the boost for the PID setpoint value.

11.15.2.5 (ID 1047) PID Setpoint Selection

Location in the menu: **P3.13.2.5**

Use this parameter to set the digital input signal that selects the PID setpoint value to be used.

11.15.2.6 (ID 332) Setpoint Source 1 Selection

Location in the menu: **P3.13.2.6**

Use this parameter to select the source of the PID setpoint signal. The AIs and the ProcessDataIn are handled as percentages (0.00–100.00%) and scaled according to the setpoint minimum and maximum.

The ProcessDataIn signals use 2 decimals.

If temperature inputs are selected, the values of parameters P3.13.1.5 Process Unit Min and P3.13.1.6 Process Unit Max must be set to correspond to the scale of the temperature measurement board: ProcessUnitMin = -50 °C and ProcessUnitMax = 200 °C.

11.15.2.7 (ID 1069) Setpoint 1 Minimum

Location in the menu: **P3.13.2.7**

Use this parameter to set the minimum value of the setpoint signal.

11.15.2.8 (ID 1070) Setpoint 1 Maximum

Location in the menu: **P3.13.2.8**

Use this parameter to set the maximum value of the setpoint signal.

11.15.2.9 (ID 1071) Setpoint 1 Boost

Location in the menu: **P3.13.2.9**

Use this parameter to set the multiplier for the setpoint boost function. When the setpoint boost command is given, the setpoint value is multiplied by the factor that is set with this parameter.

11.15.3 Feedback

11.15.3.1 (ID 333) Feedback Function

Location in the menu: **P3.13.3.1**

Use this parameter to select if the feedback value is taken from a single signal or combined from two signals. It is possible to select the mathematical function that is used when the two feedback signals are combined.

11.15.3.2 (ID 1058) Feedback Function Gain

Location in the menu: **P3.13.3.2**

Use this parameter to adjust the gain of the feedback signal. This parameter is used, for example, with the value 2 in Feedback Function.

11.15.3.3 (ID 334) Feedback 1 Source Selection

Location in the menu: **P3.13.3.3**

Use this parameter to select the source of the PID feedback signal. The AIs and the ProcessDataIn are handled as percentages (0.00–100.00%) and scaled according to the feedback minimum and maximum.

The ProcessDataIn signals use 2 decimals.

If temperature inputs are selected, the values of parameters P3.13.1.5 Process Unit Min and P3.13.1.6 Process Unit Max must be set to correspond to the scale of the temperature measurement board: ProcessUnitMin = -50 °C and ProcessUnitMax = 200 °C.

11.15.3.4 (ID 336) Feedback 1 Minimum

Location in the menu: **P3.13.3.4**

Use this parameter to set the minimum value of the feedback signal.

11.15.3.5 (ID 337) Feedback 1 Maximum

Location in the menu: **P3.13.3.5**

Use this parameter to set the maximum value of the feedback signal.

11.15.4 Feedforward

11.15.4.1 (ID 1059) Feedforward Function

Location in the menu: **P3.13.4.1**

Use this parameter to select if the feedforward value is taken from a single signal or combined from two signals. It is possible to select the mathematical function that is used when the two feedforward signals are combined.

Accurate process models are necessary for the Feedforward function. In some conditions, a gain and offset type of feedforward is sufficient. The feedforward part does not use the feedback measurements of the actual controlled process value. The feedforward control uses other measurements that affect the controlled process value.

Example

It is possible to control the water level of a tank with flow control. The target water level is set as a setpoint, and the actual level as feedback. The control signal monitors the flow that comes in.

The outflow is like a disturbance that can be measured. With the measurements of the disturbance, it is possible to try to adjust this disturbance with a feedforward control (gain and offset) added to the PID output. The PID controller reacts much faster to changes in the outflow than if the level is only measured.

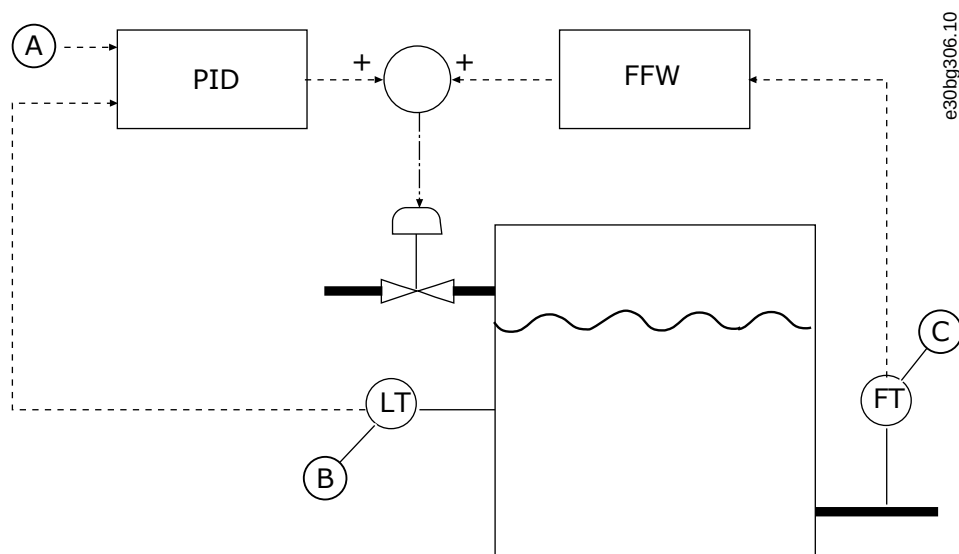


Illustration 77: The Feedforward Control

A	Level ref	C	Outflow control
B	Level control		

11.15.4.2 (ID 1060) Feedforward Gain

Location in the menu: **P3.13.4.2**

Use this parameter to adjust the gain of the feedforward signal.

11.15.4.3 (ID 1061) Feedforward 1 Source Selection

Location in the menu: **P3.13.4.3**

Use this parameter to select the source of the PID feedforward signal.

11.15.4.4 (ID 1062) Feedforward 1 Minimum

Location in the menu: **P3.13.4.4**

Use this parameter to set the minimum value of the feedforward signal.

11.15.4.5 (ID 1063) Feedforward 1 Maximum

Location in the menu: **P3.13.4.5**

Use this parameter to set the maximum value of the feedforward signal.

11.15.5 Sleep Function

11.15.5.1 (ID 1016) SP1 Sleep Frequency

Location in the menu: **P3.13.5.1**

Use this parameter to set the limit below which the output frequency of the drive must stay for a set time before the drive goes to the sleep state.

The value of this parameter is used when the signal of the PID controller setpoint is taken from the setpoint source 1.

Criteria for going to sleep mode:

- Output frequency remains below sleep frequency for longer than defined sleep delay time.
- PID feedback signal remains above defined wake-up level.

Criteria for waking from sleep

- PID feedback signal falls below defined wake-up level.

NOTICE

An incorrectly set wake-up level can prevent the drive from going into sleep mode.

11.15.5.2 (ID 1017) SP1 Sleep Delay

Location in the menu: **P3.13.5.2**

Use this parameter to set the minimum duration that the output frequency of the drive must stay below the set limit before the drive goes to the sleep state. The value of this parameter is used when the signal of the PID controller setpoint is taken from the setpoint source 1.

11.15.5.3 (ID 1018) SP1 Wake-Up Level

Location in the menu: **P3.13.5.3**

Use this parameter to set the level at which the drive wakes up from the sleep state. When the PID feedback value goes below the level that is set with this parameter, drive wakes up from the sleep state. The operation of this parameter is selected with wake-up mode parameter.

11.15.5.4 (ID 1019) SP1 Wake-Up Mode

Location in the menu: **P3.13.5.4**

Use this parameter to select the operation for the wake-up level parameter. The drive wakes up from the sleep mode when the value of PID Feedback goes below the Wake-up level.

This parameter defines if Wake-up level is used as a static absolute level or as a relative level which follows PID setpoint value.

Selection number	Selection name	Description
0	Absolute level	The wake-up level is a static level that does not follow the setpoint value.
1	Relative setpoint	The wake-up level is an offset below the actual setpoint value. The wake-up level follows the actual setpoint.

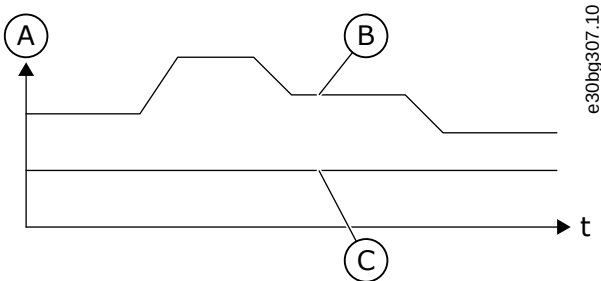


Illustration 78: Wake-up Mode: Absolute Level

A	Setpoint	C	Wake-up level
B	PID Setpoint		

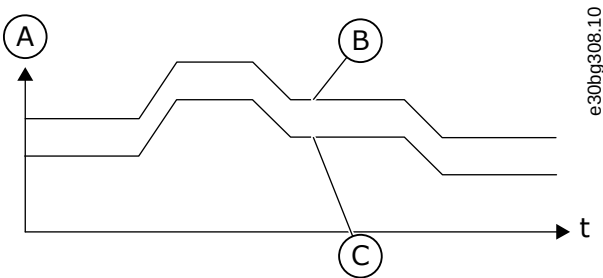


Illustration 79: Wake-up Mode: Relative Setpoint

A	Setpoint	C	Wake-up level
B	PID Setpoint		

11.15.5.5 (ID 1793) SP1 Sleep Boost

Location in the menu: **P3.13.5.5**

Use this parameter to set the value that is added to the actual setpoint value when the sleep boost function is used. Before the drive goes to the sleep state, the PID regulation setpoint increases automatically, which gives a higher process value. The sleep state is longer, also when there is some moderate leakage.

The boost level is used when there is a frequency threshold and delay, and the drive goes to the sleep state. After the increment in the setpoint by the actual value, the boost increment of the setpoint is erased and the drive goes to the sleep state and the motor stops. The boost increment is positive with the direct PID regulation (P3.13.1.8 = Normal) and negative with the reverse PID regulation (P3.13.1.8 = Inverted).

If the actual value does not go to the increment setpoint, the boost value is erased after the time set with P3.13.5.5. The drive goes to the normal regulation with the normal setpoint.

In a Multi-pump setup, if an auxiliary pump starts during the boost, the boost sequence stops and the normal regulation continues.

11.15.5.6 (ID 1795) SP1 Sleep Boost Maximum Time

Location in the menu: **P3.13.5.6**

Use this parameter to set the timeout time for the sleep boost function.

11.15.5.7 (ID 1075) SP2 Sleep Frequency

Location in the menu: **P3.13.5.5**

Use this parameter to set the limit below which the output frequency of the drive must stay for a set time before the drive goes to the sleep state.

The value of this parameter is used when the signal of the PID controller setpoint is taken from the setpoint source 1.

Criteria for going to sleep mode:

- Output frequency remains below sleep frequency for longer than defined sleep delay time.
- PID feedback signal remains above defined wake-up level.

Criteria for waking from sleep

- PID feedback signal falls below defined wake-up level.

NOTICE

An incorrectly set wake-up level can prevent the drive from going into sleep mode.

11.15.5.8 (ID 1076) SP2 Sleep Delay

Location in the menu: **P3.13.5.6**

Use this parameter to set the minimum duration that the output frequency of the drive must stay below the set limit before the drive goes to the sleep state. The value of this parameter is used when the signal of the PID controller setpoint is taken from the setpoint source 1.

11.15.5.9 (ID 1077) SP2 Wake-Up Level

Location in the menu: **P3.13.5.7**

Use this parameter to set the level at which the drive wakes up from the sleep state. When the PID feedback value goes below the level that is set with this parameter, drive wakes up from the sleep state. The operation of this parameter is selected with wake-up mode parameter.

11.15.5.10 (ID 1020) SP2 Wake-Up Mode

Location in the menu: **P3.13.5.8**

Use this parameter to select the operation for the wake-up level parameter. The drive wakes up from the sleep mode when the value of PID Feedback goes below the Wake-up level.

This parameter defines if Wake-up level is used as a static absolute level or as a relative level which follows PID setpoint value.

Selection number	Selection name	Description
0	Absolute level	The wake-up level is a static level that does not follow the setpoint value.
1	Relative setpoint	The wake-up level is an offset below the actual setpoint value. The wake-up level follows the actual setpoint.

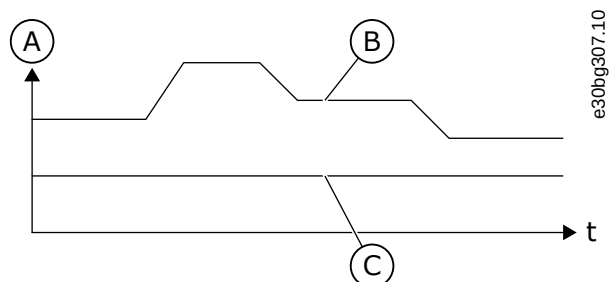


Illustration 80: Wake-up Mode: Absolute Level

A	Setpoint	C	Wake-up level
B	PID Setpoint		

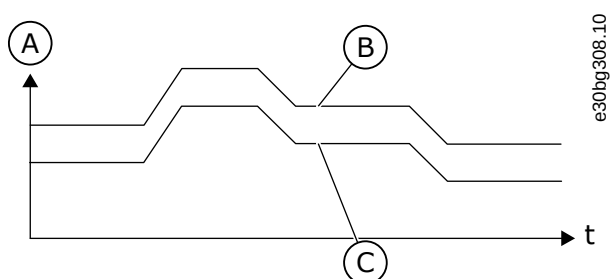


Illustration 81: Wake-up Mode: Relative Setpoint

A	Setpoint	C	Wake-up level
B	PID Setpoint		

11.15.5.11 (ID 1794) SP2 Sleep Boost

Location in the menu: **P3.13.5.11**

Use this parameter to set the value that is added to the actual setpoint value when the sleep boost function is used.

11.15.5.12 (ID 1796) SP2 Sleep Boost Maximum Time

Location in the menu: **P3.13.5.12**

Use this parameter to set the timeout time for the sleep boost function.

11.15.6 Feedback Supervision

Use the feedback supervision to make sure that the PID Feedback value (the process value or the actual value) stays in the set limits. Use this function, for example, to find a pipe break and stop the flooding.

These parameters set the range in which the PID Feedback signal stays in correct conditions. If the PID Feedback signal does not stay in the range, and this continues longer than the delay, a Feedback supervision fault (the fault code 101) shows.

11.15.6.1 (ID 735) Enable Feedback Supervision

Location in the menu: **P3.13.6.1**